Joint Learning for Feature Reconstruction and Prediction in Agricultural Semantic Segmentation From Incomplete Satellite Image Time Series

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***Abstract*—Satellite image time series (SITS) is crucial for semantic segmentation (SS) in agriculture applications. However, cloud contamination introduces time gaps in SITS, disrupting temporal dependencies and causing feature shifts, which decreases the performance of models pretrained on complete SITS. Existing methods typically address this challenge by reconstructing the entire SITS before prediction or using data augmentation (DA) to simulate missing data. Yet, full reconstruction may introduce noise and redundancy, whereas the DA model can only handle limited missing patterns, leading to poor generalization. We propose a joint learning framework with feature reconstruction (FR) and prediction to address incomplete SITS more effectively. Specifically, we simulate temporal missing scenarios using temporal masks. The two tasks are guided by both ground-truth labels and the teacher model trained on complete SITS. The prediction task constrains the model from selectively reconstructing temporal features, ensuring they align with the teacher's complete representations and are beneficial for prediction. It reduces unnecessary reconstruction and limits noise propagation. By integrating reconstructed features into the prediction task, the model avoids learning shortcuts and maintains its ability to handle varied missing patterns and complete SITS. The experiments show strong generalization capabilities in multiple aspects: 1) different SS tasks, including cropland extraction and crop classification; 2) different study areas, including Hunan Province, Western France, and Catalonia; 3) different sensors, primarily Sentinel-2 and PlanetScope; 4) different numbers of time gaps, while retaining the model's performance on complete SITS; and 5) different model architectures, including seven typical backbones. The source codes are available at: https://github.com/wangyuze-csu/Joint_FRP**

***Index Terms*—Agriculture semantic segmentation (SS) tasks, feature reconstruction (FR), incomplete satellite image time series (SITS), joint learning.**

I. Introduction

In RECENT years, deep learning (DL) methods applied to remote sensing (RS) images have played an important role in automatic agricultural monitoring tasks. Among them, semantic segmentation (SS) has shown significant potential in applications such as cropland extraction and crop classification [1], [2], [3]. These applications play a critical role in supporting agricultural resource management [4], policy planning [5], and sustainable development [6]. With the increasing availability of high temporal resolution RS images, dense satellite image time series (SITS) have become a valuable source of information for agricultural SS tasks [7], [8]. The detailed temporal dynamics captured by SITS allow DL models to effectively learn and exploit the phenological patterns of different vegetation [8]. This temporal information enhances the class separability not only between different crop types but also between cropland and other land covers [9].

However, in practical application scenarios, ensuring the completeness and quality of collected SITS remains challenging due to frequent cloud contamination. According to the International Satellite Cloud Climatology Project-Flux Data (ISCCP-FD), the global monthly cloud cover is approximately 66.5% [10], [11]. Such pervasive cloud presence often results in missing temporal observations, which not only disrupts the temporal dependencies across time steps but also induces a shift in the overall distribution of temporal information. Consequently, models trained on complete SITS often struggle to generalize to incomplete sequences, as they fail to capture the underlying phenological dynamics present in real-world data [12], [13]. This impairs the model's ability to leverage temporal information for accurate and robust prediction.

For images with minor cloud contamination, such as thin clouds or small areas of thick cloud cover, cloud removal methods [14], [15] can successfully reconstruct the affected areas, thereby preserving the temporal completeness of SITS. However, persistent and severe cloud contamination presents significant challenges to these methods. Methods that depend on multitemporal images [16], typically require the availability of cloud-free reference images within a relevant temporal window, which is often impractical under such conditions. Alternative approaches attempt to reconstruct cloud-covered regions using spatial and spectral information from cloud-free areas within a single image [17]. Yet, when cloud coverage exceeds a critical threshold (e.g., >50% coverage), the significant loss of contextual information makes it hard for these methods to recover the missing areas accurately. These heavily or fully cloud-obscured images introduce time gaps in SITS. Although satellites with shorter revisit intervals can

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help avoid cloud contamination, time gaps are inevitable in cloud-prone regions. This is because clouds may persist in such regions, making it difficult to find suitable time windows even with high-frequency data. As a result, maintaining the performance of DL models with incomplete SITS has become a critical and unresolved challenge in agricultural applications.

Recently, two primary strategies have been developed to address the challenges posed by incomplete SITS: data reconstruction (DR) and data augmentation (DA). DR methods aim to reconstruct the SITS before the input into multitemporal SS models (hereafter referred to as the prediction models) [18]. One category of DR methods makes use of the external information from other optical or synthetic aperture radar (SAR) sensors to fill the time gaps [19]. However, acquiring cloud-free optical images within the relevant temporal window is often impractical due to the persistent presence of cloud cover. The acquisition and processing of SAR data increase time and manual costs [20]. Furthermore, due to the varying cross-modality mapping relationships across different locations and years, integrating SAR and incomplete optical SITS under low-cost remains a challenge [21]. Another type of DR method is using the internal information from the incomplete STIS itself. These methods typically rely on interpolation [22] or DL-based methods [23]. Although the interpolation might perform well when the time gaps are sparse or isolated, it fails in cases of extensive or continuous time gaps [22], [24]. DL-based reconstruction can offer improved results, but may still contain noise and distort temporal consistency [25], [26]. More critically, complete reconstruction is redundant for prediction tasks. Prediction models primarily depend on key temporal patterns rather than the entire time series [27], [28]. Thus, complete reconstruction might not only involve unnecessary computational and storage costs, but also introduce the risk of overfitting to noise and irrelevant information [29].

The DA methods aim to enhance the model's robustness to incomplete SITS by increasing the diversity of training data through simulated temporal missing conditions [30]. In the temporal domain, two widely adopted strategies in RS are window slicing (WS) and temporal dropping (TD) [31]. WS augments the training data by sampling continuous local temporal segments, enabling models to infer results using localized phenological cues [32]. TD, on the other hand, randomly removes time steps across the full sequence, enhancing the model's ability to handle globally discontinuous temporal patterns. Although these strategies are simple and effective, they face limitations. These include the risk of catastrophic forgetting [33], where the model loses its ability to process complete SITS, and mutual interference when learning from diverse missing patterns [34]. Moreover, models trained under simulated temporal missing conditions may suffer from the shortest-path phenomenon [35], where they learn shortcut reasoning strategies rather than generalizable temporal representations. The model tends to rely on the shortest and computationally least expensive temporal jump connections, while neglecting more complete but longer temporal dependency chains [35]. It will compromise the model's generalizability to unseen data conditions.

To address these problems, we proposed a joint learning framework with feature reconstruction (FR) and prediction for agricultural segmentation tasks using SITS. It enables the model to selectively reconstruct essential temporal features that are beneficial for prediction under temporal missing conditions. Specifically, we simulate incomplete SITS by randomly masking time steps during training. The model is jointly guided by both prediction labels and a teacher model trained on complete SITS. The FR task encourages the model to generate temporal features from masked inputs that approximate those extracted by the teacher model from complete SITS. It allows the model to allocate its capacity to recovering global temporal changing patterns rather than local details, which simplifies the reconstruction task and reduces the risk of introducing errors. Simultaneously, supervision from prediction labels ensures that the model learns to recover task-relevant temporal features rather than reconstructing irrelevant or noisy information. The more concise representations also help the model focus on informative features instead of redundant details during prediction. For the prediction tasks, the reconstructed temporal feature can support the reasoning process. By explicitly guiding the reasoning flow from the reconstructed features (start) to the logit outputs (intermediate) and finally to the labels (end), the model is prevented from taking shortcuts that overly focus on missing patterns and rely solely on local cues. This design enables the model to develop robust and complete reasoning capabilities based on long-term temporal dependencies, allowing it to generalize across different missing conditions while still performing effectively on complete SITS.

In summary, FR prevents the model from acquiring incorrect reasoning ability during prediction, whereas prediction maintains reliability and reduces the redundancy of reconstructed information. To validate our framework, we conduct experiments on cropland extraction and crop classification tasks with SITS. It contains three study areas in Hunan Province and Western France and Catalonia, using images from Sentinel-2 and PlanetScope satellites, respectively. We also assessed the generalization ability of the proposed framework in varied model architectures and investigated its internal mechanisms. In summary, our main contributions are as follows.

1) We proposed a joint learning framework with FR and prediction for the agricultural SS task using SITS in temporal missing conditions. It outperforms the state-of-the-art method in cropland extraction and crop classification tasks.
2) The proposed framework demonstrates strong generalization across different sensors, regions, and varied numbers of time gaps, while being agnostic in various model backbones. It significantly improves model robustness for real-world agricultural monitoring.

The remainder of this article is organized as follows. Section II reviews related work on DR and DA methods. Section III describes the data processing pipeline and the proposed framework in detail. Section IV presents the experimental setup, comparative results with other methods, and the model's performance under different numbers of time gaps. Section V discusses the model-agnostic nature and internal

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mechanisms of the proposed approach. Finally, Section VI concludes the article and outlines directions for future work.

II. Related Work

A. DR for Time Gaps in SITS

DR aims to impute the time gaps in SITS, thereby providing complete temporal information to support the SS models in prediction tasks [18]. Depending on the source of information, these methods can be broadly categorized into external and internal reconstructions. The external reconstruction typically fills the data gaps by utilizing the information from other optical or SAR sensors [36], [37]. According to the varied revisit cycles of different satellites, the optical-based method uses adjacent available images from other satellites to complete the SITS [36]. However, acquiring such images within a relevant temporal window is often impractical, particularly in regions with persistent cloud cover during rainy seasons, such as tropical, monsoon, and maritime climates. Furthermore, these periods often correspond to the key growth stages of multiple crops [38], making it difficult to fill in key temporal information. As a result, it is hard to effectively support multitemporal cropland extraction or crop classification.

SAR images have the ability to penetrate clouds, making them a reliable source of information in cloudy scenarios. Nevertheless, SAR data acquisition and preprocessing can be time-consuming [39]. The instability of SAR imaging systems and the presence of speckle noise reduce visual quality [40], which can compromise the texture and boundary information in reconstructed data. This may further introduce noise that negatively affects the delineation of cropland and crop boundaries in multitemporal agricultural SS [41]. Moreover, due to the distinct imaging mechanisms of optical and SAR images, different statistical properties, temporal and semantic patterns make it challenging to directly use SAR to fill time gaps in optical SITS. Meanwhile, cropland extraction and crop classification often rely on subtle spectral differences with temporal changes [42], which are difficult to recover through SAR-driven reconstruction due to the fundamental disparity between optical reflectance and SAR backscattering mechanism. Although [21]and [37] learn the mapping cross-modality relationships between SAR and optical data to jointly construct complete time information. Considering the diverse characteristics of optical and SAR images that are exhibited in different locations and across years [43], they still need extensive paired optical-SAR samples to relearn the cross-modality relationships for different scenarios. It still increased the computational costs and data requirements.

To address the cost issue, the internal reconstruction methods typically leverage available temporal information within the SITS to fill the time gaps [18]. Among them, interpolation estimates missing values under the assumption of temporal continuity, using neighboring time points as references [22]. For instance, linear interpolation (Linear-IN), closest interpolation (Closest-IN), and last interpolation (Last-IN) are widely used in the RS field. However, these methods have limited practical value in agriculture because they ignore correlations between different time steps, including both trends and seasonal variations across different crops [44]. When faced with long or continuous temporal gaps, the limited contextual information often leads to substantial deviations from true values [22]. With the advancement of DL methods, data-driven models can learn the changing pattern of temporal features to fill the time gaps in incomplete SITS [23]. These methods are usually trained on simulated incomplete SITS, with the optimization goal of generating data that closely approximates complete SITS. For example, generative adversarial networks (GANs) [45] employ an adversarial strategy between the generator and the discriminator, which forces the generators to estimate the time gaps that are indistinguishable from complete SITS by the discriminator. Time-series vision transformer (TSVit) [46] captures global temporal dependencies through self-attention mechanisms and utilizes cross-attention to select information relevant to the missing information, thereby reconstructing the time gaps. In addition, the integration of convolutional and pooling operations with attention mechanisms enables more comprehensive extraction of multiscale local temporal features, enhancing the modeling of global temporal dynamics [47]. Despite these advances, reconstructing large or continuous gaps remains challenging. The severe lack of contextual information may cause the reconstructed data to deviate from the overall distribution of SITS, thereby introducing noise [25], [26]. The propagation of such noises may further affect the reasoning process of prediction models, resulting in unreliable segmentation results.

In summary, the external reconstruction methods are limited by the availability of optical data and the cost of using SAR data. The internal reconstruction methods are unsuitable for handling the extensive or continuous time gaps in SITS. Even with the support of DL reconstruction methods, it still faces the risk of noise propagation. More importantly, fully reconstructing all missing information is often unnecessary for multitemporal SS tasks. Although the reasoning process of the model's prediction jointly relies on spatial and temporal features, spatial information in SITS is often redundant [27]. In the temporal dimension, the model mainly relies on the key temporal patterns extracted from SITS [26]. Reconstructing the entire sequence may introduce irrelevant information, increasing computational/storage demands and risking overfitting to noise or nonessential features.

B. DA for Time Gaps in SITS

DA is widely used in the RS field to enhance the robustness of models in complex scenarios by manually increasing the diversity of training data. For the multitemporal segmentation tasks, exposing the model to scenarios with missing temporal data during training has proven effective for improving its ability to handle time gaps during prediction [31]. Among them, the most commonly used methods are WS and TD [31], [48]. The WS strategy refers to selecting a temporal window (i.e., a continuous sequence of time steps) from the original SITS to simulate scenarios with missing data [33]. WS encourages the model to focus on local temporal features and to infer predictions based on partial sequences. However, due to the different life cycles of various crops [49], WS may completely remove key phenological stages, forcing the

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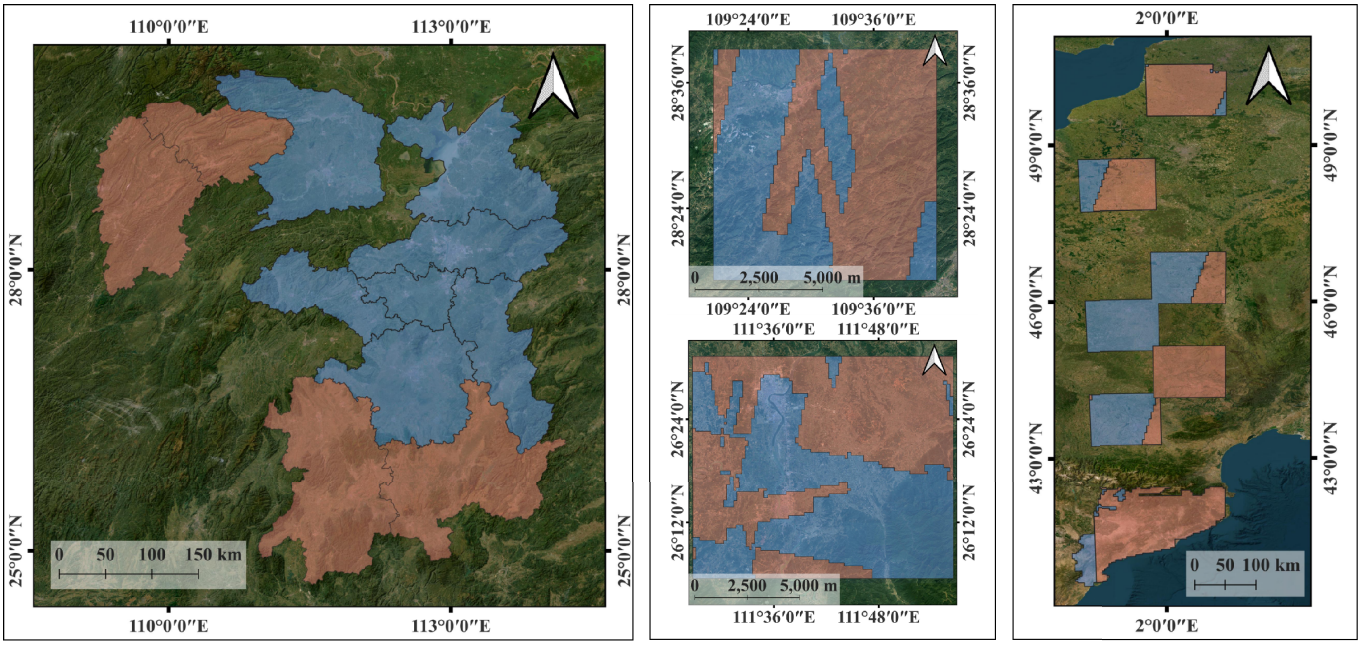

Fig. 1. Location and extent of datasets, where red and blue regions represent complete and incomplete subsets, respectively. The Hunan SEN and Hunan PLA datasets were collected from the Sentinel-2 and Planet platforms, and are used for cropland extraction. The Fr&Cat S4A data are derived from an existing dataset and used for crop classification. (a) Hunan SEN. (b) Hunan PLA. (c) Fr&Cat S4A.

model to rely solely on local information and resulting in underfitting of the global temporal dynamics during training. TP is a more general strategy that typically removes time steps either individually or continuously during training to simulate temporal missing scenarios [50]. It enables the model to infer complete temporal features from partially observed data, thereby reducing the model's reliance on patterns in complete SITS. Furthermore, the TP can prevent the model from over-relying on specific time steps and guide it to learn more robust and generalized temporal features. For models handling incomplete SITS, customizing TD strategies based on local crop phenology calendars and climatic conditions, and subsequently deploying multiple customized models, is considered beneficial [51]. However, agricultural systems are highly heterogeneous across different regions, with different crops and climate zones exhibiting distinct temporal–spectral change patterns [49]. Meanwhile, cloud contamination is affected by multiple factors—including atmospheric conditions, topography, and temperature [52]. The influence of those multiple factors makes subdividing regions and customizing TD strategies inefficient [53]. In addition, while cloud contamination often follows predictable seasonal trends, it also contains inherent randomness. The fixed TP strategy may cause the model to lose its ability to handle exceptional cases. For example, in northeastern Hunan, although Sentinel-2 data show a high cloud-related missing rate in summer and autumn (averaging 50%), nonnegligible gaps also occur in spring and winter (averaging 15%). Therefore, the random TP strategy became an efficient and robust strategy in the RS field.

Although the DA is an effective way to enhance the model's ability to handle the time gaps in the multitemporal segmentation task, it still faces several challenges. First, due to catastrophic forgetting [33], models with limited memory capacity may overfit incomplete SITS, thereby diminishing their ability to process complete SITS effectively [34]. Moreover, training on multiple temporal missing patterns may result in mutual interference, making it difficult for the model to generalize across varied missing conditions. Second, during the prediction of segmentation tasks, the model may fail to develop appropriate reasoning capabilities due to the shortcut learning phenomenon [35]. The model tends to take the easier and computationally least expensive way to predict the final result. The model may overlook the intermediate dynamic modeling process within the full temporal dependency chain and instead adopt jump connections for reasoning. For instance, the model may learn to infer crop types directly by relying only on observations from January and April in simulated temporal missing scenarios, while ignoring the intermediate growth stages. As a result, in a real-world scenario, even when data from February and March becomes available, the model continues to follow this shortcut path instead of incorporating it into a complete temporal reasoning chain. Once the model develops a reliance on such shortcut behavior, it tends to learn the correlation between missing patterns and the final prediction result, rather than building a global representation from limited observations. This not only impairs the model's ability to capture long-term temporal dependencies, but also degrades its generalization to unseen data.

III. Materials and Methods

A. Preprocessing and Datasets

Our study areas for dataset construction include Hunan Province in China and two European regions: Western France

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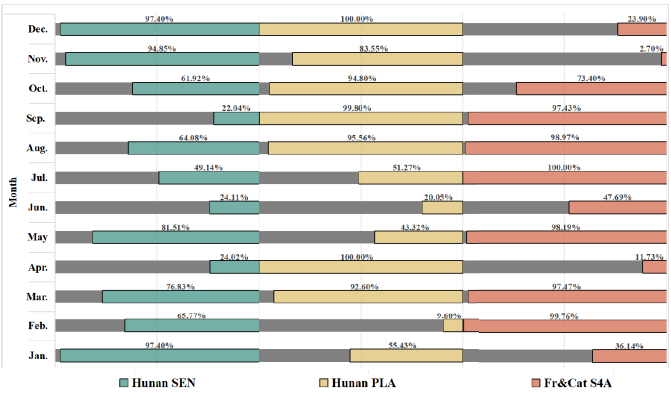

Fig. 2. Monthly data availability of SITS samples in incomplete subsets: Hunan SEN, Hunan PLA, and Fr&Cat S4A datasets. For each dataset, we counted the number of available images for each month and calculated the monthly data availability as the proportion of available images relative to the total number of samples for that month.

and Catalonia. Hunan Province has a continental subtropical monsoonal humid climate, while the European regions have temperate maritime and Mediterranean climates. All study areas have frequent and persistent cloud cover, which might lead to extensive and continuous time gaps in the acquired SITS. For China's Hunan Province, we prepared cropland extraction datasets using two sensors with different spatial and temporal resolutions: Sentinel-2 and PlanetScope constellation, resulting in the Hunan SEN and Hunan PLA datasets, respectively. For the European regions, we constructed the Fr&Cat S4A dataset based on the S4A for crop classification [54]. We further divided each dataset into complete and incomplete subsets based on the temporal availability of SITS. The complete subsets (60% for training and 40% for testing) were used for simulated experiments, and the incomplete subsets were only used for real-world testing. Detailed information and processing steps are presented below.

1) Hunan SEN: We collected Sentinel-2A and Sentinel-2B Level-2A Bottom of Atmosphere reflectance images (S2 L2A) from Google Cloud Platform, covering the period from January 2019 to December 2019. Bands with 10- and 20-m spatial resolutions were used for dataset construction, with the 20-m bands resampled to 10-m resolution. Furthermore, we used the quality assurance (QA) bands provided by the ESA on Google Earth Engine (GEE) as a reference to calculate the cloud covers [55].

As shown in Fig. 1(a), for the complete subset, we randomly sampled 256 × 256 pixel subregions across seven city-level administrative areas of northeastern Hunan, where cloud-cover frequency was relatively low. For each subregion, cloud-covered pixels were excluded to generate monthly median composite images from all available pixels. Composite images with cloud-cover ratios <20% are treated as clean images and used to assemble SITS. After filtering, 5000 SITS samples with complete 12-month images were obtained. For the incomplete subset, we randomly sampled 256 × 256 pixel subregions across four city-level administrative areas of northwestern and southeastern Hunan, where cloud-cover frequency was relatively high. Monthly composite images with cloud cover >40% were treated as missing. The remaining cloud-cover pixels were filled using Linear-IN from the nearest available observations. After filtering, 3048 SITS samples with 4–9 available monthly images were obtained. The monthly image availability for incomplete samples is shown in Fig. 2. For all samples, the labels are obtained by visually interpreting the RGB bands of Sentinel-2 images, assisted by high-resolution Google images.

2) Hunan PLA: We acquired PlanetScope-2 (PS2) images from the Planet Platform, covering the period from January 2019 to December 2019. They include blue, green, red, and near-infrared spectral bands with a spatial resolution of 3.7 m. As shown in Fig. 1(b), the regions are located in the Jishou and Yongzhou areas. During the downloading, we selected three images with the lowest cloud-cover ratios for the target area, prioritizing an even distribution throughout each month. We also manually masked the cloud-cover regions for each image and generated monthly median composite images.

All images were cropped to 256 × 256 pixel images, and subimages with >40% cloud-covered ratios were treated as missing. The remaining cloud-cover pixels were filled using Linear-IN from the nearest available observations. After assembling monthly images into SITS samples, we divide them into complete and incomplete subsets according to the available time length. The complete subset contained 5184 samples with 10–12 monthly images, where samples missing 1–2 monthly images were gap-filled using Linear-IN from the nearest available images. The incomplete subset contained 4584 samples with 4–9 available monthly images. The monthly image availability for incomplete samples is shown in Fig. 2. For all samples, the labels are obtained by visually interpreting the RGB bands of PS2 images.

3) Fr&Cat S4A: Based on S4A datasets, we extracted data from January to December 2019, including 10 spectral bands at both 10- and 20-m spatial resolutions. The 20-m resolution bands were resampled to 10 m to match the other bands. Each SITS sample covered a 366 × 366 pixel area, with the cloud-cover ratio of every image calculated using the provided Hollstein masks. Images with >40% cloud-cover pixels were treated as missing, followed by additional manual filtering of remaining unusable images. We created month median composites with the remaining invalid pixels filled via Linear-IN from the nearest valid observations.

The complete subset obtained 4771 samples with 10–12 monthly images, where samples missing 1–2 monthly images were gap-filled using Linear-IN from the nearest available images. The incomplete subset contained 2925 samples with 4–9 monthly images. The monthly image availability for incomplete samples is shown in Fig. 2. For all samples, we combined the S4A labels into eight categories: background (BG), cereals (Cer), vegetables and fruits (V&F), root and tuber crops (R&T), leguminous crops (LC), oilseed crops (OC), sugar crops (SC), and other crops (OT).

B. Method

Multitemporal SS models are designed to process time-series inputs and produce a single semantic output, widely applied in agricultural tasks such as cropland extraction and

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crop classification. The input SITS is typically represented as $X \in \mathbb{R}^{T\times C\times H\times W}$, and the model extracts spatial–temporal features to generate a pixel-wise one-hot prediction $Y \in \mathbb{R}^{H\times W\times K}$ with K classes.

As shown in Fig. 3, our proposed framework aims to jointly optimize the model through FR and prediction tasks under simulated data-missing scenarios, thereby enhancing its ability to handle incomplete SITS. These two tasks are simultaneously constrained by the teacher model pretrained in complete SITS and the ground truth labels.

1) In the FR task, the teacher model serves as a reference to demonstrate to the student model what the ideal spatial–temporal features extracted from complete SITS should look like. Accordingly, the student model is encouraged to reconstruct features from incomplete SITS that align with the ideal features. Meanwhile, ground truth labels also encourage the model to selectively reconstruct the key patterns of features that are beneficial for predictions, which balances the trade-off between reconstructed feature completeness and prediction reliability.
2) In the prediction task, supported by these reconstructed features, the student model was encouraged to imitate the teacher's decision-making process. The model is supervised by both the teacher's logit outputs (from complete SITS) and ground truth labels, enabling it to learn the intrinsic interclass relationships and underlying reasoning logic for robust feature representation. This strategy prevents the student model from overfitting to limited temporal missing patterns and maintains its ability to generalize to complete sequences. The detailed workflow is illustrated as follows.

First, we conduct teacher model pretraining and simulate data-missing scenarios. We define a training dataset $\mathcal{D} = \{X_i\}_{i=1}^{N}$, where each sample $X_i \in \mathbb{R}^{T\times C\times H\times W}$ represents a complete SITS. The teacher model f_T is pretrained on $\mathcal{D}$ and is frozen afterward for guiding the student model. For every sample $X \in \mathcal{D}$, we generate an incomplete version $X' \in \mathbb{R}^{T'\times C\times H\times W}$ (where $T' < T$) through random temporal masking

$$\mathbf{X}' = \text{TemporalMask}(\mathbf{X}, r), \quad r \sim \mathcal{U}(M, N) \tag{1}$$

where M and N denote the minimum and maximum mask ratios in the temporal dimension, respectively. Considering real-world temporal missing rates, M and N are typically set to 25

Second, we guide the student model f_S to selectively reconstruct critical features from Incomplete SITS. The X and X' are, respectively, fed into the f_T and f_S to produce spatial–temporal features F and F'. Considering the redundancy of spatial information in SITS, we use the deepest temporally fused feature as a supervision target. This encourages the student model to focus on reconstructing global temporal dependencies and dynamics, rather than spatial patterns. For instance, in U-TAE [56], the spatial feature from the last layer of the encoder is fused with the temporal attention. In 3-D CNN [57], the fused feature is used after processing by the last encoder layer. For RNN-based models [58], the final hidden state is used. In TSVit [59], the used features are derived by extracting class token embeddings from the temporal transformer output.

Furthermore, we standardize the obtained features F and F' to Z and Z'. As shown in Fig. 4, this is because the temporal missing of samples can cause a global shift of the feature's distribution. Standardization helps the student model avoid paying excessive attention to the values and boundaries of the features, and instead pay more attention to the key patterns of features. Specifically, F and F' are standardized along the channel dimensional, then the f_S is trained by minimizing the squared $L2$ distance between Z and Z'

$$Z = \frac{F - \mu_F}{\sigma_F}, \quad Z' = \frac{F' - \mu_{F'}}{\sigma_{F'}} \tag{2}$$

$$\mathcal{L}_{\text{FR}} = \frac{1}{N}\sum_{i=1}^{N} \|Z_i - Z'_i\|_2^2 \tag{3}$$

where $\mu_{\mathbf{F}}/\mu_{\mathbf{F}'}$ and $\sigma_{\mathbf{F}}/\sigma_{\mathbf{F}'}$ are the mean and standard deviation of $\mathbf{F}$ and $\mathbf{F}'$ along the channel dimension.

Finally, in the prediction task, we not only expect the model to obtain the correct one-hot prediction, but also guide the model to mimic the teacher model's reasoning process. On the one hand, after inputting X', the prediction result Y' of f_S are generated with support of F', which is supervised by ground truth label Y

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{C} Y_{i,k}\log\left(Y'_{i,k}\right). \tag{4}$$

It also constrains the f_S to reconstruct the critical patterns of F', which is beneficial for prediction. On the other hand, by inputting X and X' to f_T and f_S, respectively, we obtained logit outputs l and l'. The f_S is trained by minimizing the Kullback–Leibler divergence between the logits, thereby enabling the f_S to learn how the f_T handles the relative relationships between different classes

$$\mathcal{L}_{\text{KD}} = \text{KL}\left(\frac{\exp(l/T)}{\sum_{k=1}^{C}\exp\left(l^k/T\right)}, \frac{\exp(l'/T)}{\sum_{k=1}^{C}\exp\left(l'^k/T\right)}\right) \tag{5}$$

where T represents the distillation temperature, and C represents the number of classes. The entire student model is jointly optimized by the overall loss $\mathcal{L}$, which is composed of $\mathcal{L}_{\text{FR}}$ from the FR task and $\mathcal{L}_{\text{KD}}/\mathcal{L}_{\text{CE}}$ from prediction tasks

$$\mathcal{L} = \sigma\mathcal{L}_{\text{FR}} + \gamma\mathcal{L}_{\text{KD}} + \lambda\mathcal{L}_{\text{CE}} \tag{6}$$

where σ, γ, and λ are the weight coefficients for the respective losses.

IV. Experiments and Results

A. Experiment Settings

The experiments utilize the three datasets introduced in Section III, Hunan SEN and Hunan PLA are used for cropland extraction, and the Fr&Cat S4A datasets are used for crop classification. Each 256×256 pixel or 366×366 pixel sample is normalized channel-wise by the mean and variance computed across all samples from the corresponding dataset. To validate the effectiveness and generalizability of the proposed method,

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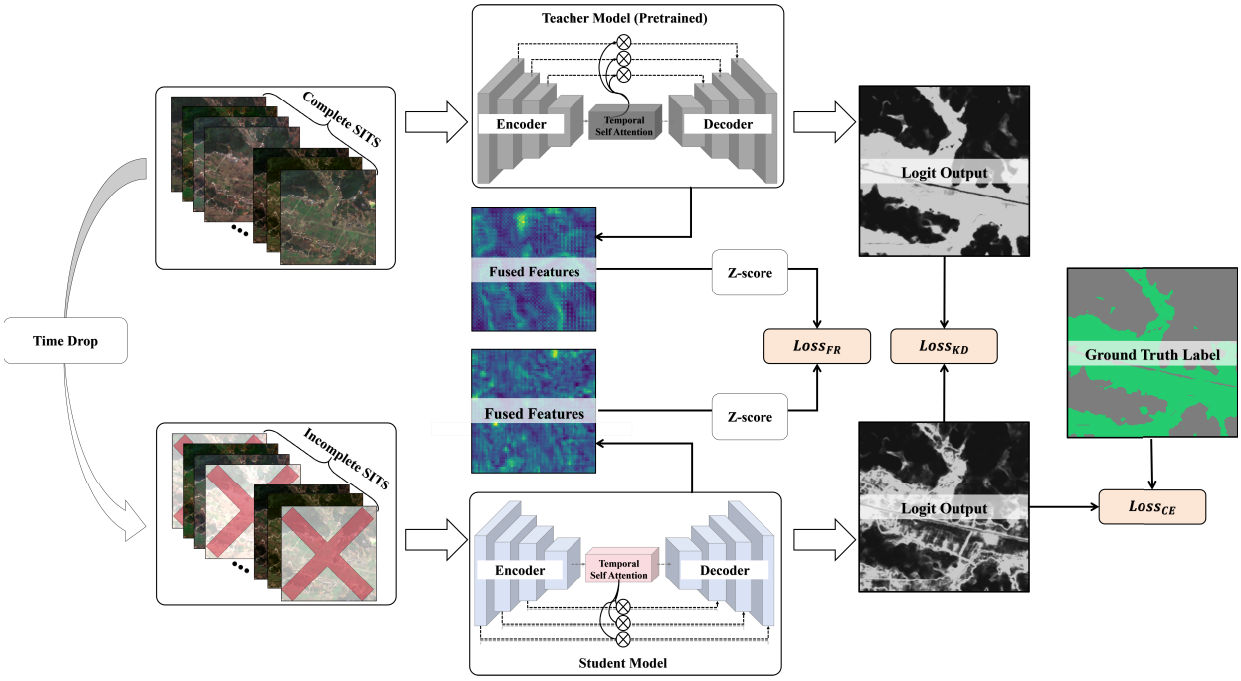

Fig. 3. General framework of the proposed method. The general framework of the proposed method. 1) In the FR task, the teacher model serves as a reference to show the student model what the ideal spatial–temporal features extracted from complete SITS should look like. Supervision is provided via standardization (Z-score) features, and the FR loss (Loss_{FR}) is computed. The teacher model has been fully trained on the complete SITS. 2) In the prediction task, supported by these reconstructed features, the student model is encouraged to imitate the teacher's decision-making process. Supervision is provided using the intermediate outputs (logits) from the teacher model and the ground truth labels, and the corresponding losses (Loss_{KD} and loss_{CE}) are computed. In the manuscript, losses will be abbreviated as $\mathcal{L}$.

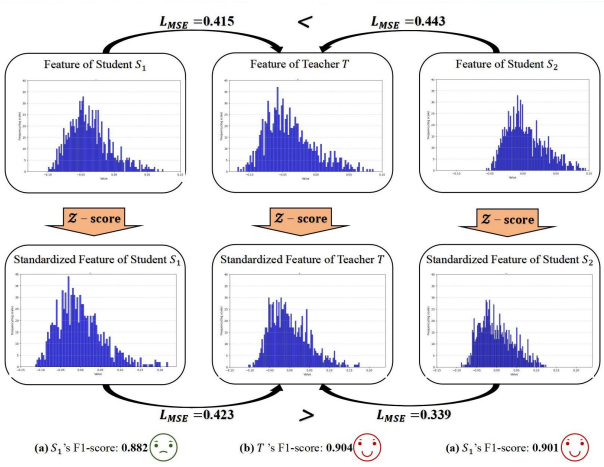

Fig. 4. Toy case where two students, S_1 and S_2, are guided by the same teacher T. Before the standardization (Z-score), S_1 reconstructed features from incomplete SITS that are more similar to the teacher's ideal features, yet it achieves a lower $F1$-score. In contrast, features from S_2 are less similar to ideal features and get a higher mean squared error distance L_{mse}, but they get a higher $F1$-score. After standardization, the issue is solved, which makes the model focus more on feature distribution instead of values and boundaries.

all experiments were conducted in both simulated and real-world environments. For simulated experiments, we dropped 25%–75% time steps of each SITS sample from complete subsets, with a 60%/40% split for training and test sets, respectively. For real-world experiments, we employed the models trained in simulated environments, testing them on all samples from the incomplete subsets. All experiments use U-TAE [56] as the backbone.

To evaluate the accuracy of our results, we employed multiple assessment metrics: overall accuracy (OA), average accuracy (AA), mean intersection over union (mIoU), Kappa, and $F1$-scores for both individual classes and all classes (mean $F1$-scores) collectively. The $F1$-score, serving as the harmonic mean of precision (user's accuracy) and recall (producer's accuracy), effectively quantifies the model's classification reliability for cropland and crop types.

All models were trained using PyTorch on an Ubuntu 16.04 operating system with an NVIDIA A6000 GPU (48-GB memory). Each model was trained using the AdamW optimizer, with a batch size of 4 and 100 epochs. The learning rate of our method was initially set to 1×10^{-3}.

B. Main Experiments

To validate the effectiveness of the proposed method for agricultural SS tasks and its generalization capability across different sensors, we conducted experiments on cropland extraction and crop classification tasks using Sentinel-2 and PlanetScope SITS. All experiments were performed on three distinct datasets: Hunan SEN (based on Sentinel-2 SITS for cropland extraction), Hunan PLA (based on PlanetScope SITS for cropland extraction), and Fr&Cat S4A (based on Sentinel-2 SITS for crop classification). As described in Section III-A, each dataset is divided into complete and incomplete subsets to construct simulated and real-world experimental environments.

1) Compared Methods: We compared the proposed method with two method categories, namely, DR and DA, for agricultural SS tasks using SITS with time gaps. The baseline method refers to training the model on 60% of the SITS samples from

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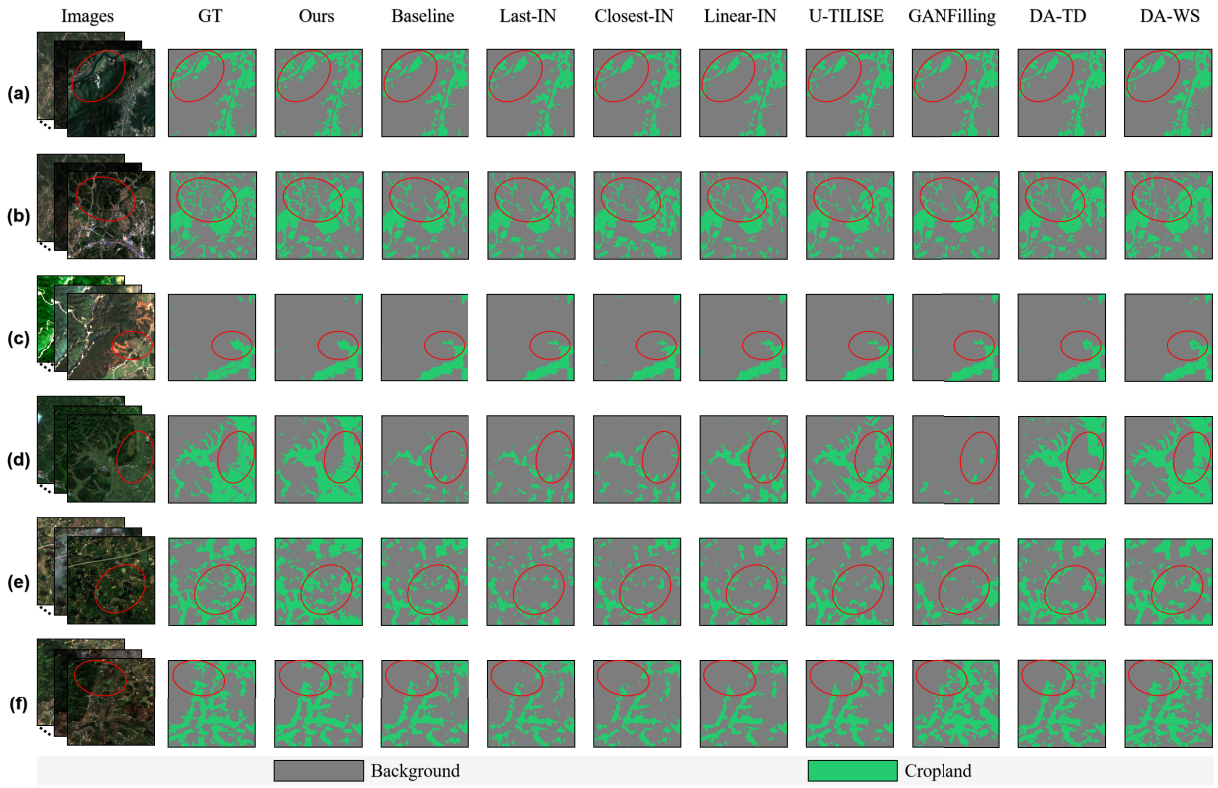

Fig. 5. Visualization of sample results for our and compared methods in the Hunan SEN dataset from: (a)–(c) simulated experiments and (d)–(f) real-world experiments.

TABLE I

PERFORMANCE OF COMPARED METHODS AND OUR METHOD IN SIMULATED EXPERIMENTS ON THE HUNAN SEN DATASET. THE BASELINE REPRESENTS THE ORIGINAL MODEL TRAINED ONLY ON THE COMPLETE SITS (CL-$F1$: CROPLAND $F1$-SCORE, BG-$F1$: BG $F1$-SCORE, AND M-$F1$: MEAN $F1$-SCORES)

Method	OA(%)	AA(%)	Kappa	mIoU(%)	M-F1(%)	CL-F1(%)	BG-F1(%)
Baseline	89.53	87.87	0.71	75.15	85.26	77.34	93.19
Last-IN	88.18	87.28	0.66	71.72	82.73	73.03	92.43
Closest-IN	88.80	87.48	0.68	73.33	83.94	75.11	92.77
Linear-IN	88.71	87.56	0.68	73.06	83.74	74.74	92.73
U-TILIES	88.38	86.69	0.67	72.62	83.44	74.39	92.49
GANFilling	88.89	86.34	0.69	74.26	84.67	76.62	92.71
DA-TD	89.57	87.88	0.71	75.25	85.34	77.47	93.21
DA-WS	89.75	87.42	0.72	76.06	85.93	78.60	93.26
Ours	90.84	88.12	0.75	78.66	87.72	81.53	93.91

the complete subsets and directly testing it on the simulated or real-world incomplete SITS without any adaptation.

a) DR methods: These methods perform comprehensive SITS reconstruction before model prediction. For direct interpolation, we implement three typical methods: Linear-IN, Closest-IN, and Last-IN. For DL-based methods, we compare two state-of-the-art approaches: UTILIZE [47] and GANFilling [45]. These DR models were retrained on the same data as our method, specifically using 60% of complete subsets from each dataset. To simulate temporal missing scenarios, we generated sample pairs by applying 25%–75% temporal dropout to construct corresponding complete/incomplete SITS samples.

b) DA methods: These methods enhance prediction model robustness in simulated temporal missing scenarios by increasing training data diversity. We implement two typical methods in the temporal dimension: WS (DA-WS) [31] and TD (DA-TD) [48]. WS randomly extracts continuous temporal windows covering 25%–75% of each SITS sample's full sequence length. TD randomly drops 25%–75% of time steps from each SITS sample. Each sample has a 50% probability of being augmented during training.

2) Cropland Extraction With Sentinel-2 SITS: As shown in Tables I and II and Fig. 5, in the simulated and real-world environment, most of the DR methods failed; they even performed worse than the Baseline. Specifically, the cropland $F1$-score decreases by 0.93%–5.57% compared to the Baseline in the simulated experiment. Except for UTILIZE, the cropland $F1$-score of other DR methods decreases by 6.87%–26.37% compared to the baseline in

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TABLE II

PERFORMANCE OF COMPARED METHODS AND OUR METHOD IN REAL-WORLD EXPERIMENTS ON THE HUNAN SEN DATASET. THE BASELINE REPRESENTS THE ORIGINAL MODEL TRAINED ONLY ON THE COMPLETE SITS (CL-$F1$: CROPLAND $F1$-SCORE, BG-$F1$: BG $F1$-SCORE, AND M-$F1$: MEAN $F1$-SCORES)

Method	OA(%)	AA(%)	Kappa	mIoU(%)	M-F1(%)	CL-F1(%)	BG-F1(%)
Baseline	85.78	83.31	0.41	57.83	69.46	47.14	91.78
Last-IN	83.87	79.80	0.28	52.07	62.75	34.71	90.80
Closest-IN	84.31	81.73	0.30	52.97	63.81	36.57	91.05
Linear-IN	85.42	83.94	0.37	56.22	67.62	43.61	91.63
U-TILIES	87.14	84.85	0.48	62.01	73.77	55.05	92.49
GANFilling	80.21	67.24	0.32	53.34	65.94	43.90	87.99
DA-TD	88.72	87.38	0.56	66.37	77.80	62.23	93.37
DA-WS	89.69	81.40	0.61	69.60	80.60	67.32	93.88
Ours	89.49	83.98	0.64	71.14	81.99	70.37	93.61

real-world experiments. Due to insufficient information support, inaccurately reconstructed images inevitably introduce noise into the entire SITS, which misleads the model and results in incorrect classification outcomes.

DA methods outperform DR methods, compared to the baseline, DA-TD, and DA-WS, achieving improvements of 0.17% and 1.63%, respectively. By exposing the model to simulated temporal missing scenarios during training, these methods enable the model to learn how to match incomplete temporal features with correct labels under various temporal missing patterns. However, they are still limited by the model's memory capacity, which hinders its ability to generalize across different missing patterns. Their generalization ability to unseen data is also affected by the shortcut learning phenomenon, where the model tends to rely on a jump connection for reasoning, rather than developing a robust reasoning capability that depends on long-term dependency. Moreover, the model may overfit on severely missing conditions (e.g., 50%–75% missing), and lose its ability to generalize to complete SITS or relatively mild missing conditions (e.g., 0%–25%). Detailed experimental results and analysis can be found in Appendix A. Unlike the similar performance of DA-WS and DA-TD in the simulated experiments, DA-WS shows an 8.18% improvement over DA-TD in cropland $F1$-score. This is due to the continuous-time gaps in real-world incomplete SITS, which are closer to the simulated training data of DA-WS. However, DA-WS does not show a significant advantage when applied to SITS collected from sensors with higher temporal resolution, and it exhibits substantial failure in crop classification tasks (details in Sections IV-B3 and IV-B4).

As shown in Tables I and II, the results in the simulated and real-world experiments demonstrate that our method outperforms the other methods. Compared to the top-performing method, it shows an improvement of 4.53% and 3.73% in the cropland $F1$-score, and surpasses the baseline by 5.42% and 49.28% in cropland $F1$-score. The proposed method not only mitigates the negative impact of noise introduced during the comprehensive reconstruction process, but also leverages the reconstructed features to provide reliable support for the model to learn more robust reasoning capabilities. Fig. 5 shows the qualitative results of our methods and the compared methods in both simulated and real-world experiments. From Fig. 5(a), (b), and (e), our method captures more fragmented and elongated croplands with smaller field sizes. In Fig. 5(c), (d), and (f), it also demonstrates higher boundary completeness and retains more details.

In addition, it can be observed that the baseline performs worse in real-world experiments compared to simulated experiments. This is also because the SITS in real-world experiments contain more continuous time gaps. The self-attention mechanism in U-TAE enables the model to dynamically allocate attentions, thereby reconstructing temporal dependencies by leveraging strongly correlated neighboring time steps. However, as the ratio of continuous time gaps increases, it becomes difficult for the model to reconstruct any information from the completely missing segments. A more detailed explanation can be found in Section V-A.

3) Cropland Extraction With PlanetScope SITS: Compared with Sentinel-2, the SITS from PlanetScope has a higher temporal resolution. This leads to a lower probability of continuous missing time steps in real-world SITS, with missing time steps tending to follow a more random distribution. This is because the availability of images increases during cloudy and rainy seasons (e.g., spring and winter in Hunan), while time gaps occur relatively more frequently in other seasons.

As shown in Tables III and IV, the performance trend of our method and the compared methods in the Hunan PLA dataset is similar to Hunan SEN. First, all DR methods still perform worse than the baseline, with cropland $F1$-score decreases ranging from 22.93% to 7.00% and 1.85% to 10.56% in the simulated and real-world experiments, respectively. Second, unlike the obvious advantage of DA-WS over DA-TD on the Hunan SEN dataset, DA-WS and DA-TD achieve similar results in real-world experiments, with cropland $F1$-scores of 79.82% and 79.45%, respectively. This demonstrates that the reduced probability of continuous missing time steps diminishes the advantage of DA-WS. Finally, our method also demonstrates strong robustness on SITS from PlanetScope in both simulated and real-world experiments. It surpassed the top-performing compared method by 2.19% and 3.46% in cropland $F1$-score, and outperformed the baseline by 8.06% and 6.01% in cropland $F1$-score. As shown in Fig. 6, our method produces cropland extraction results with higher completeness and finer detail compared to other methods.

For instance, in Fig. 6(a) and (e), our method shows cropland boundaries that more closely resemble the labels.

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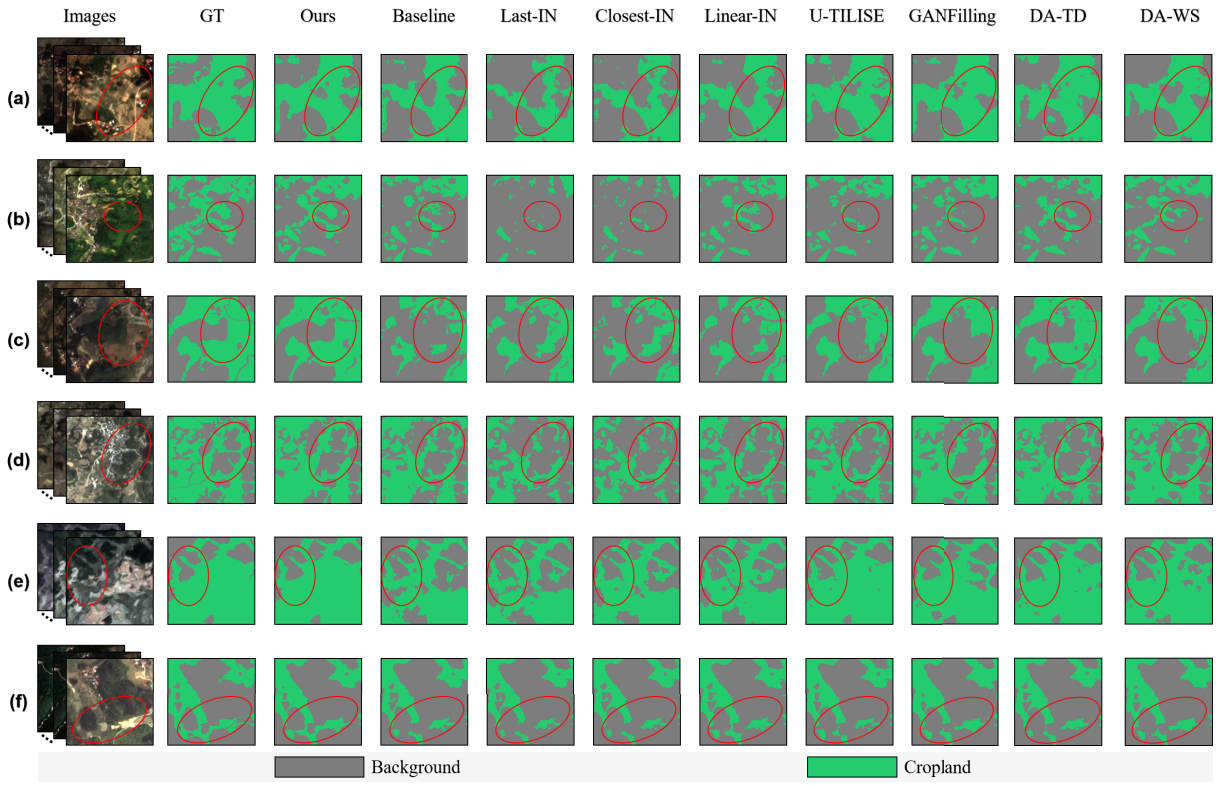

Fig. 6. Visualization of sample results for our and compared methods in the Hunan PLA dataset from: (a)–(c) simulated experiments and (d)–(f) real-world experiments.

TABLE III

PERFORMANCE OF COMPARED METHODS AND OUR METHOD IN SIMULATED EXPERIMENTS ON THE HUNAN PLA DATASET. THE BASELINE REPRESENTS THE ORIGINAL MODEL TRAINED ONLY ON THE COMPLETE SITS (CL-$F1$: CROPLAND $F1$-SCORE, BG-$F1$: BG $F1$-SCORE, AND M-$F1$: MEAN $F1$-SCORES)

Method	OA(%)	AA(%)	Kappa	mIoU(%)	M-F1(%)	CL-F1(%)	BG-F1(%)
Baseline	87.87	87.22	0.67	72.34	83.30	74.56	92.04
Last-IN	85.60	86.04	0.59	66.96	79.09	67.42	90.75
Closest-IN	85.71	86.31	0.59	67.13	79.22	67.61	90.83
Linear-IN	86.29	86.82	0.61	68.42	80.26	69.34	91.17
U-TILIES	86.27	86.72	0.61	68.41	80.25	57.46	96.94
GANFilling	86.25	84.94	0.62	69.21	80.96	62.07	95.20
DA-TD	88.94	86.50	0.71	75.57	85.67	78.84	92.51
DA-WS	88.76	86.17	0.71	75.32	85.51	78.64	92.37
Ours	89.97	88.14	0.74	77.40	86.90	80.57	93.24

Fig. 6(c) and (d) demonstrates more internal details within croplands as well as additional small cropland fields. Meanwhile, our method achieved a more accurate extraction of fields exhibiting unique phenological patterns that differ from other cropland fields within the same scene. For instance, in Fig. 6(b) and (f), our method performs better on certain croplands with distinctive phenological patterns.

4) Crop Classification With Sentinel-2 SITS: As shown in Tables V and VI, compared with cropland extraction tasks, the performance of our method and the compared methods show different results in crop classification tasks. The DA methods no longer outperform DR methods, and most of the compared methods exhibit worse performance compared with the baseline. Specifically, in the simulated experiments, compared to the baseline, only DA-TD showed an increase in the mean $F1$-scores by 0.54%, while the mean $F1$-scores of all other methods decreased by 1.37%–20.09%. In the real-world experiments, all compared methods exhibited a decrease in mean $F1$-scores ranging from 6.05% to 42.45%.

Specifically, for the crops with a single growing season and distinct temporal pattern (e.g., Cer, OC, and SC), most compared methods achieve higher performance in both simulated and real-world experiments. The underperformance of the WC method on OC is mainly attributed to OC's relatively short key growing period, which makes its distinctive temporal features easily masked during training. For crops with more complex temporal patterns, indicating multiple growth cycles, such as R&T, LC, and OT, most comparison methods perform poorly. The DR methods may focus more on reconstructing local details while underfitting the broader global temporal patterns.

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TABLE IV

PERFORMANCE OF COMPARED METHODS AND OUR METHOD IN REAL-WORLD EXPERIMENTS ON THE HUNAN PLA DATASET. THE BASELINE REPRESENTS THE ORIGINAL MODEL TRAINED ONLY ON THE COMPLETE SITS (CL-$F1$: CROPLAND $F1$-SCORE, BG-$F1$: BG $F1$-SCORE, AND M-$F1$: MEAN $F1$-SCORES)

Method	OA(%)	AA(%)	Kappa	mIoU(%)	M-F1(%)	CL-F1(%)	BG-F1(%)
Baseline	90.60	88.53	0.72	76.27	85.97	77.90	94.03
Last-IN	90.02	88.02	0.70	74.81	84.92	76.16	93.69
Closest-IN	89.97	88.10	0.70	74.63	84.79	75.91	93.67
Linear-IN	90.10	88.72	0.70	74.72	84.84	75.91	93.77
U-TILIES	90.30	88.49	0.71	75.40	85.34	69.67	96.49
GANFilling	89.92	87.25	0.70	74.92	85.02	76.46	93.58
DA-TD	90.41	86.08	0.74	77.30	86.77	79.82	93.71
DA-WS	90.38	86.24	0.73	77.04	86.59	79.45	93.72
Ours	90.67	88.34	0.76	79.17	88.10	82.58	93.63

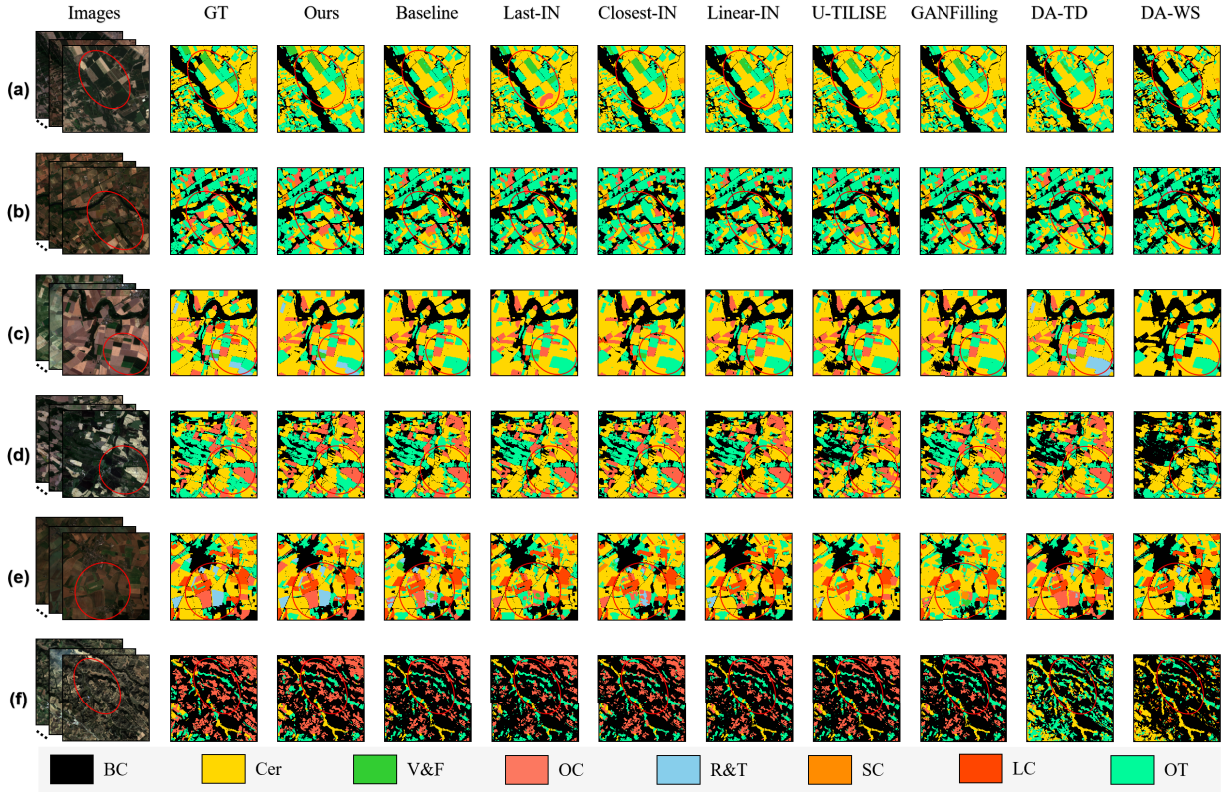

Fig. 7. Visualization of sample results for our and compared methods in Fr&Cat S4A dataset from: (a)–(c) simulated experiments and (d)–(f) real-world experiments.

For DA methods, due to the limited memory capacity, models that rely mainly on shortcut reasoning struggle to effectively handle crops with complex temporal patterns under diverse missing conditions. For the crops with short growing seasons and no distinct temporal patterns (e.g., V&F), all methods fail to achieve effective reconstruction or inference for this class, due to the lack of sufficient informative temporal signals. In addition to these crop-specific temporal characteristics, environmental and climatic variability across different climate regions may further influence model performance. Variations in temperature regimes, precipitation patterns, and growing-season lengths can alter crop growth dynamics and the distinctness of their temporal signatures, thereby affecting both reconstruction and prediction accuracy. These environmental differences may also exacerbate the challenges encountered for crops with complex or weak temporal profiles, particularly under missing-data conditions (the detailed NDVI profiles can be found in https://github.com/wangyuze-csu/Joint_FRP).

Our method still outperforms all other compared methods. In simulated and real-world experiments, it achieved an improvement of 9.76% and 10.52% in mean $F1$-scores compared to the best-performing methods. It also achieves an improvement of 10.35% and 3.83% in the mean $F1$-scores compared to the baseline. The results demonstrate that our method also has strong generalization capability across different agricultural SS tasks. As shown in Fig. 7, in both

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TABLE V

PERFORMANCE OF COMPARED METHODS AND OUR METHOD IN SIMULATED EXPERIMENTS ON THE FR&CAT S4A DATASET. THE BASELINE REPRESENTS THE ORIGINAL MODEL TRAINED ONLY ON THE COMPLETE SITS (M-$F1$: MEAN $F1$-SCORES)

Method	OA(%)	AA(%)	Kappa	mIoU(%)	M-F1(%)
Baseline	85.10	78.10	0.77	59.79	72.46
Last-IN	84.01	84.26	0.76	57.36	69.96
Closest-IN	84.76	86.29	0.77	59.02	71.47
Linear-IN	84.38	82.00	0.76	56.65	69.45
U-TILIES	81.87	72.66	0.73	48.42	60.93
GANFilling	78.62	72.20	0.67	43.50	57.90
DA-TD	85.34	86.17	0.78	60.75	72.85
DA-WS	80.49	60.26	0.70	45.34	55.32
Ours	88.90	83.71	0.83	69.16	79.96

TABLE VI

PERFORMANCE OF COMPARED METHODS AND OUR METHOD IN REAL-WORLD EXPERIMENTS ON THE FR&CAT S4A DATASET. THE BASELINE REPRESENTS THE ORIGINAL MODEL TRAINED ONLY ON THE COMPLETE SITS (M-$F1$: MEAN $F1$-SCORES)

Method	OA(%)	AA(%)	Kappa	mIoU(%)	M-F1(%)
Baseline	84.45	70.83	0.76	54.84	67.59
Last-IN	82.78	79.77	0.74	50.85	62.48
Closest-IN	82.85	80.54	0.74	51.80	63.50
Linear-IN	82.87	76.94	0.74	48.03	59.66
U-TILIES	79.62	49.88	0.69	33.82	44.14
GANFilling	82.29	67.45	0.73	48.12	60.61
DA-TD	83.10	79.00	0.74	50.71	62.56
DA-WS	76.10	41.73	0.62	30.01	38.90
Ours	85.10	74.32	0.77	57.89	70.18

simulated and real-world experiments, our method is able to accurately identify the boundaries between different crop fields and exhibits robustness in complex mixed-cropping scenarios. For instance, our method achieves more accurate boundary delineation for V&F and Cer. In Fig. 7(c) and (e), it successfully identifies multiple crop types in complex scenes, including OC, OT, SC, and Cer. In Fig. 7(d) and (f), it provides more complete boundary extraction for fields with Cer and OT.

However, our method also has limitations. As shown in Tables VII and VIII, the V&F and LC achieve lower accuracy compared to OT, especially in real-world experimental environments, with only 21.36% and 47.28% in $F1$-score, respectively. The main reasons are as follows.

a) *High intraclass variability of V&F and LC:* It prevents the teacher model from extracting accurate temporal features and learning correct reasoning pathways, even when using complete SITS. As a result, the supervision signals contain substantial noise, which negatively affects model performance.

b) *Short growth cycles and no distinct temporal patterns of V&F:* The large amount of time gaps in SITS causes most of V&F's distinguishable features to be completely lost, making it difficult for the proposed method to reconstruct the temporal features (the detailed analysis is provided in Section IV-C).

c) *Intercrop similarity of LC and OT:* It not only makes it difficult for the teacher model to learn discriminative features and provide incorrect supervision signals, but also causes the trained model to reconstruct erroneous features when the clearly distinctive parts are completely masked (The detailed intraclass variability and confusion metrics can be found at https://github.com/wangyuze-csu/Joint_FRP).

5) Time and Computational Efficiency: The comparison table listing the computational and time cost metrics during the training and inference (testing) phases is shown in Table IX. Specifically, the evaluation metrics include time per epoch and peak GPU memory during training, while time per item and FLOPs are considered during inference. As illustrated in Table IX, the additional computational and time costs are reported relative to the baseline model, which is trained on complete SITS. The experiments were conducted on the Hunan SEN simulated dataset, which was used as a representative case.

For DR methods, interpolation-based approaches are non-trainable and can be directly applied. They incur no additional training cost and no extra computational cost during inference, but they increase the inference time. In contrast, DL-based methods significantly raise both computational and time costs during training and inference. For DA methods, the computational and time costs are slightly increased during training due to extra data preprocessing. During the inference, the model architecture remains unchanged, so there is no additional computational or time cost.

For the proposed method, although it introduces additional computational and time costs during training, these costs are generally lower than those of DL-based DR methods, while achieving better performance than all DR methods. Moreover, similar to DA methods, the proposed method does not change the model architecture, resulting in no extra cost during inference, yet it still improves performance compared to DA methods. Due to the extra computational and time cost during the training phase, the proposed method is not necessary in the ideal scenario where the SITS is completely clean. However, obtaining entirely cloud-free SITS over cloud-prone regions is often difficult during the mapping process, and complete and incomplete SITS typically co-exist in the acquired data in most regions. This demonstrates that the proposed method achieves a good balance between cost and performance, thus showing strong practical applicability.

C. Ablation Experiments

Although Section IV-B has demonstrated the effectiveness of our proposed method, its robustness under varying levels of temporal incompleteness remains unclear. Therefore, we further investigate the effectiveness and limitations of our method by controlling the temporal missing rates in a simulated experiment environment. Specifically, we set temporal missing rates from 0% to 75%. This allows us to evaluate the model's performance on complete SITS and its robustness under low (25%), middle (50%), and high (75%) levels of temporal incompleteness.

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TABLE VII

*F*1-SCORES (%) OF EACH CLASS FOR COMPARED AND OUR METHOD IN SIMULATED EXPERIMENTS ON THE FR&CAT S4A DATASET, WHERE BC, CER, V&F, OC, R&T, LC, SC, AND OT DENOTE BACKGROUND, CEREALS, VEGETABLES AND FRUITS, OILSEED CROPS, ROOT AND TUBER CROPS, LEGUMINOUS CROPS, SUGAR CROPS, AND OTHER CROPS, RESPECTIVELY. THE BASELINE REPRESENTS THE ORIGINAL MODEL TRAINED ONLY ON THE COMPLETE SITS

Method	BC	Cer	V&F	OC	R&T	LC	SC	OT
Baseline	90.88	86.11	38.38	77.28	79.20	45.42	87.62	74.82
Last-IN	90.43	84.67	31.08	79.40	73.65	39.96	87.89	72.60
Closest-IN	90.67	85.79	34.95	80.50	74.44	41.86	89.43	74.15
Linear-IN	90.60	85.17	31.92	79.81	67.64	40.19	86.26	73.98
U-TILIES	90.02	80.57	12.92	65.77	52.32	29.36	85.05	71.40
GANFilling	87.48	77.95	23.18	60.43	53.58	33.37	61.40	65.83
DA-TD	90.91	86.28	35.16	79.99	79.36	44.43	91.24	75.46
DA-WS	87.44	84.59	0.00	0.00	73.50	42.98	83.99	70.02
Ours	92.57	90.87	51.19	86.59	87.60	56.02	93.11	81.74

TABLE VIII

*F*1-SCORES (%) OF EACH CLASS FOR COMPARED AND OUR METHOD IN REAL-WORLD EXPERIMENTS ON THE FR&CAT S4A DATASET, WHERE BC, CER, V&F, OC, R&T, LC, SC, AND OT DENOTE BACKGROUND, CEREALS, VEGETABLES AND FRUITS, OILSEED CROPS, ROOT AND TUBER CROPS, LEGUMINOUS CROPS, SUGAR CROPS, AND OTHER CROPS, RESPECTIVELY. THE BASELINE REPRESENTS THE ORIGINAL MODEL TRAINED ONLY ON THE COMPLETE SITS

Method	BC	Cer	V&F	OC	R&T	LC	SC	OT
Baseline	88.65	88.21	18.93	78.53	68.82	44.75	74.50	78.36
Last-IN	88.11	83.06	6.06	78.23	59.60	25.50	83.03	76.23
Closest-IN	88.13	83.17	7.85	78.25	65.33	25.87	82.98	76.38
Linear-IN	88.21	82.79	9.16	78.22	40.14	23.01	78.98	76.77
U-TILIES	87.99	79.09	8.86	51.99	11.12	20.67	19.48	73.91
GANFilling	87.94	83.62	18.14	77.17	23.67	44.58	74.74	75.03
DA-TD	88.32	83.08	8.74	78.69	56.24	26.80	81.58	77.02
DA-WS	83.90	84.26	0.00	0.00	15.47	32.84	26.64	68.07
Ours	88.80	89.20	21.36	81.52	74.09	47.28	79.55	79.6

TABLE IX

ADDITIONAL COMPUTATIONAL AND TIME COSTS OF THE COMPARED METHODS AND OUR METHOD RELATIVE TO THE BASELINE DURING TRAINING AND INFERENCE PHASES IN THE SIMULATED EXPERIMENT ON THE HUNAN SEN DATASET

Method	Trainable	Training Time/epoch (s)	Peak GPU Memory(GB)	GFLOPs	Inference Time(ms/img)	mIou(%)
Baseline	✓	760.1	11.36	150.12	0.05	75.15
...+Last-IN	×	/	/	...+0.00	...+0.10	71.72
...+Closet-IN	×	/	/	...+0.00	...+0.40	73.33
...+Linear-IN	×	/	/	...+0.00	...+0.39	73.06
...+U-TILIES	✓	...+1045.66	...+23.00	...+498.46	...+2.31	72.62
...+GANFilling	✓	...+3330.12	...+5.51	...+482.02	...+3.11	74.26
...+DA-TD	✓	...+18.54	...+0.65	...+0.00	...+0.00	75.25
...+DA-WS	✓	...+20.57	...+0.66	...+0.00	...+0.00	76.06
...+Ours	✓	...+582.01	...+9.95	...+0.00	...+0.00	78.66

As shown in Tables X and XI, in cropland extraction tasks, our method maintains strong classification capability in complete time series and exhibits good generalization across different levels of temporal missing rates. First, compared with the baseline on the complete SITS (0%), our method achieves improvements of 1.67% and 1.09% in the cropland *F*1-score with the Hunan SEN and Hunan PLA datasets. It benefits from the ability to reconstruct features of the model and the self-distillation process [51]. The reconstructed features can support the model to avoid over-rely on shortcut reasoning paths. Knowledge distillation acts as a regularizing function during the model's relearning process by penalizing high variation in the local feature space, thereby mitigating the impact of overfitting [60]. Second, under low (25%) and middle (50%) levels of temporal missing conditions, our method achieves similar or increased performance compared with the baseline model's performance on complete SITS (0%). It maintains a cropland *F*1-score of over 81.26% and 80.97% on Hunan SEN and Hunan PLA, respectively. This indicates that our method can reconstruct key temporal features from limited temporal information, which supports the model to make accurate predictions. Third, our method also faces

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TABLE X

PERFORMANCE OF THE BASELINE AND OUR METHOD UNDER DIFFERENT TEMPORAL MISSING RATES ON THE HUNAN SEN DATASET. THE BASELINE REPRESENTS THE ORIGINAL MODEL TRAINED ONLY ON THE COMPLETE SITS (CL-$F1$: CROPLAND $F1$-SCORE, BG-$F1$: BG $F1$-SCORE, AND M-$F1$: MEAN $F1$-SCORES)

Mask ratio	Methods	OA(%)	AA(%)	Kappa	mIoU(%)	M-F1(%)	CL-F1(%)	BG-F1(%)
0%	Baseline	90.69	88.53	0.75	78.06	87.29	80.72	93.86
	Ours	91.14	88.64	0.76	79.24	88.09	82.07	94.12
25%	Baseline	90.33	88.41	0.73	77.13	86.66	79.65	93.66
	Ours	91.04	88.54	0.76	79.01	87.94	81.83	94.05
50%	Baseline	89.47	87.87	0.70	75.03	85.18	77.20	93.15
	Ours	90.80	88.32	0.75	78.47	87.58	81.26	93.90
75%	Baseline	86.33	85.51	0.59	67.52	79.43	67.53	91.34
	Ours	90.54	89.13	0.74	77.32	86.77	79.70	93.83

TABLE XI

PERFORMANCE OF THE BASELINE AND OUR METHOD UNDER DIFFERENT TEMPORAL MISSING RATES ON THE HUNAN PLA DATASET. THE BASELINE REPRESENTS THE ORIGINAL MODEL TRAINED ONLY ON THE COMPLETE SITS (CL-$F1$: CROPLAND $F1$-SCORE, BG-$F1$: BG $F1$-SCORE, AND M-$F1$: MEAN $F1$-SCORES)

Mask ratio	Methods	OA(%)	AA(%)	Kappa	mIoU(%)	M-F1(%)	CL-F1(%)	BG-F1(%)
0%	Baseline	90.47	88.66	0.75	78.48	87.63	81.69	93.56
	Ours	90.67	88.34	0.76	79.17	88.10	82.58	93.63
25%	Baseline	90.01	88.40	0.74	77.38	86.88	80.48	93.29
	Ours	90.55	88.29	0.76	78.88	87.91	82.26	93.56
50%	Baseline	87.85	87.37	0.67	72.21	83.19	74.35	92.04
	Ours	90.13	88.25	0.74	77.76	87.15	80.97	93.33
75%	Baseline	79.87	83.74	0.35	53.18	65.87	44.02	87.73
	Ours	87.31	87.46	0.65	70.78	82.10	72.44	91.76

TABLE XII

PERFORMANCE OF THE BASELINE AND OUR METHOD UNDER DIFFERENT TEMPORAL MISSING RATES ON THE FR&CAT S4A DATASET. THE BASELINE REPRESENTS THE ORIGINAL MODEL TRAINED ONLY ON THE COMPLETE SITS (M-$F1$: MEAN $F1$-SCORES)

Mask ratio	Methods	OA(%)	AA(%)	Kappa	mIoU(%)	M-F1(%)
0%	Baseline	90.27	87.98	0.85	75.21	84.77
	Ours	89.96	86.79	0.85	74.27	84.04
25%	Baseline	89.32	86.10	0.84	71.63	82.05
	Ours	89.75	86.36	0.85	73.23	83.19
50%	Baseline	85.24	77.94	0.77	59.28	72.12
	Ours	89.02	84.08	0.84	69.86	80.47
75%	Baseline	72.89	61.59	0.57	35.17	48.50
	Ours	85.91	76.14	0.79	58.21	70.70

challenges under a high level (75%) of missing conditions, it only achieves cropland $F1$-scores of 79.70% and 72.44% on Hunan SEN and Hunan PLA, respectively. The relatively poorer performance on the Hunan PLA is due to the increased spatial resolution, which demands more refined reconstructed features for the model's prediction. However, our method is more focused on reconstructing the global temporal features rather than restoring detailed information. In addition, the performance of DA methods is presented in Appendix A. Although the model performs well in middle (50%) and high (75%) level missing conditions, it is prone to overfitting in these conditions, and loses its ability to handle complete SITS and low-level (25%) missing conditions.

As shown in Table XII, in the crop classification task, the performance of our method decreased with the increase of temporal missing rates. But it still keeps the mean $F1$-scores of 80.47% under complete SITS, low (25%), and middle (50%) level missing conditions. According to the $F1$-score of each class presented in Table XIII, the overall decline is mainly attributed to the performance drop on V&F and LC. When the temporal missing rate is 0%–50%, the $F1$-score for other classes only decreased from 0.19% to 1.56% compared to the baseline model's performance on complete SITS (0%). Specifically, the significant drop in V&F is mainly because the increased temporal missing ratio may result complete absence of their growth cycles. Our method requires temporal anchor points to effectively recover temporal patterns. For LC, the decline is mainly due to the high intraclass variability, which prevents the teacher model from effectively learning discriminative features. Consequently, the transfer of incorrect

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TABLE XIII

$F1$-SCORES (%) OF THE BASELINE AND OUR METHOD UNDER DIFFERENT TEMPORAL MISSING RATES ON THE FR&CAT S4A DATASET. THE BASELINE REPRESENTS THE ORIGINAL MODEL TRAINED ONLY ON THE COMPLETE SITS (BC: BACKGROUND, CER: CEREALS, V&F: VEGETABLES AND FRUITS, OC: OILSEED CROPS, R&T: ROOT AND TUBER CROPS, LC: LEGUMINOUS CROPS, SC: SUGAR CROPS, AND OT: OTHER CROPS)

Mask ratio	Methods	BC	Cer	V&F	OC	R&T	LC	SC	OT
0%	Baseline	93.03	92.46	66.23	88.55	93.93	64.89	95.51	83.61
	Ours	92.85	92.27	62.97	88.31	93.21	64.25	95.15	83.34
25%	Baseline	92.74	91.32	59.34	86.93	90.58	59.26	94.45	81.73
	Ours	92.79	91.99	59.62	87.97	92.53	62.60	94.98	83.03
50%	Baseline	91.27	86.32	42.16	77.25	76.45	42.08	86.65	74.80
	Ours	92.60	90.93	51.84	86.63	89.17	56.87	93.80	81.93
75%	Baseline	84.39	66.94	15.07	42.81	37.04	17.57	66.65	57.50
	Ours	91.74	86.41	35.01	79.49	73.74	37.19	84.38	77.62

TABLE XIV

EVALUATION OF THE PROPOSED METHOD UNDER DIFFERENT NETWORKS IN SIMULATED EXPERIMENTS ON THE HUNAN SEN DATASET (CL-$F1$: CROPLAND $F1$-SCORE, BG-$F1$: BG $F1$-SCORE, AND M-$F1$: MEAN $F1$-SCORES). THE BASELINE REPRESENTS THE ORIGINAL MODEL TRAINED ONLY ON THE COMPLETE SITS

Model	Method	OA(%)	AA(%)	Kappa	mIoU(%)	M-F1(%)	CL-F1(%)	BG-F1(%)
ConvLSTM	Baseline	85.21	83.59	0.56	65.39	77.70	64.77	90.64
	Ours	90.50	88.12	0.74	77.74	87.09	80.45	93.73
ConvGRU	Baseline	86.70	86.30	0.60	68.16	79.93	68.28	91.58
	Ours	90.48	88.29	0.74	77.60	86.99	80.24	93.73
UConvLSTM	Baseline	90.36	88.18	0.74	77.29	86.77	79.89	93.66
	Ours	90.80	88.28	0.75	78.50	87.60	81.31	93.90
FPN-ConvLSTM	Baseline	81.08	74.99	0.48	60.76	74.17	60.81	87.53
	Ours	90.68	88.22	0.75	78.19	87.39	80.96	93.83
3D U-Net	Baseline	72.09	66.30	0.35	51.83	67.02	54.09	79.95
	Ours	90.22	87.64	0.73	77.22	86.74	79.95	93.53
U-TAE	Baseline	89.53	87.87	0.71	75.15	85.26	77.34	93.19
	Ours	90.84	88.12	0.75	78.66	87.72	81.53	93.91
TSViT	Baseline	89.31	87.42	0.70	74.66	84.92	76.77	93.06
	Ours	90.45	87.91	0.74	77.72	87.08	80.48	93.68

knowledge during distillation may mislead the learning process of the student model. Under high-level missing conditions, our method still maintains the mean $F1$-scores of 70.70%. Although it fails to recognize V&F and LC, with $F1$-scores of only 35.01% and 37.19%, respectively, it achieves $F1$-scores ranging from 73.74% to 91.74% for the remaining classes. In summary, the proposed method demonstrates limitations in the crop with short growth cycles and the limited by the teacher model's classification ability. However, it can still reconstruct most of the key temporal features under varying degrees of temporal missing conditions in SITS, thereby supporting the model's reasoning process for most classes.

V. DISCUSSION

A. Network Agnostic Framework

To evaluate the generalizability of the proposed framework across different network architectures in temporal missing conditions, we conducted experiments on typical types of network architectures: 1) recurrent neural networks (RNNs-based): four backbones are used, including convolutional long short-term memory (ConvLSTM) [58], convolutional gated recurrent unit (ConvGRU) [61], U-Net with ConvLSTM (U-ConvLSTM) [62], and feature pyramid network with ConvLSTM (FPN-ConvLSTM) [63]. 2) 3-D convolutional networks: we employ the 3-D U-Net [57]. 3) Self-attention-based networks: two backbones are utilized, including the temporo-spatial vision transformer (TSViT) [59] and U-Net with temporal attention encoder (U-TAE) [56]. All experiments were conducted in the Hunan SEN dataset, involving both simulated and real-world scenarios on the complete and incomplete subsets. The baseline refers to the model trained in complete SITS samples from complete subsets and directly tested in simulated and real-world incomplete SITS.

As shown in Tables XIV and XV, the proposed method shows improvement across all backbones in both simulated and real-world experiments. Specifically, in simulated experiments, our method maintains the Cropland $F1$-score above 80% for most backbones. For RNN-based networks, it achieves improvements ranging from 1.78% to 33.14%. On the 3-D U-Net, the cropland $F1$-score increases by 47.81%. For self-attention-based networks, improvements range from 4.83% to 5.42%. In real-world experiments, which involve more continuous time gaps, our method is still able to maintain the Cropland $F1$-score above 70% for most backbones. On RNN-based networks, the Cropland $F1$-score improves by 2.00%–43.04%. For the 3-D U-Net, the improvement reaches 25.42%, while for self-attention-based networks, it ranges

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TABLE XV

EVALUATION OF THE PROPOSED METHOD UNDER DIFFERENT NETWORKS IN REAL-WORLD EXPERIMENTS ON THE HUNAN SEN DATASET (CL-$F1$: CROPLAND $F1$-SCORE, BG-$F1$: BG $F1$-SCORE, AND M-$F1$: MEAN $F1$-SCORES). THE BASELINE REPRESENTS THE ORIGINAL MODEL TRAINED ONLY ON THE COMPLETE SITS

Model	Method	OA(%)	AA(%)	Kappa	mIoU(%)	M-F1(%)	CL-F1(%)	BG-F1(%)
ConvLSTM	Baseline	84.07	74.63	0.53	63.80	76.45	63.06	89.84
	Ours	89.23	82.61	0.65	71.57	82.41	71.46	93.36
ConvGRU	Baseline	86.08	77.31	0.58	66.93	78.95	66.69	91.20
	Ours	89.00	83.08	0.62	70.09	81.18	69.05	93.31
UConvLSTM	Baseline	90.70	87.27	0.67	73.22	83.52	72.65	94.40
	Ours	91.18	88.18	0.69	74.38	84.39	74.10	94.68
FPN-ConvLSTM	Baseline	80.77	69.04	0.39	56.34	69.42	50.79	88.05
	Ours	89.79	83.69	0.66	72.62	83.19	72.65	93.73
3D U-Net	Baseline	84.31	74.06	0.48	61.48	74.02	57.68	90.37
	Ours	90.44	85.48	0.67	72.88	83.29	72.35	94.22
U-TAE	Baseline	85.78	83.31	0.41	57.83	69.46	47.14	91.78
	Ours	89.49	83.98	0.64	71.14	81.99	70.37	93.61
TSViT	Baseline	83.10	72.57	0.45	59.82	72.60	55.64	89.56
	Ours	89.14	82.48	0.65	71.88	82.68	72.11	93.26

from 29.06% to 49.28%. The network-agnostic ability of the proposed framework allows it to be flexibly adapted to a variety of real-world scenarios. It can be easily integrated into existing agricultural monitoring systems without requiring structural modifications, especially in environments with computational constraints or model-specific limitations.

We further analyzed the distinct performance trends of different backbones in simulated and real-world experiments. The key findings are summarized as follows.

1) The baseline performance of 3-D convolutional networks is relatively low in both simulated and real-world experiments. This is mainly because their spatiotemporal modeling is embedded in the most fundamental units of the network [64]. As a result, they struggle to properly align the temporal dimension when inputs are incomplete. This limitation not only restricts its ability to capture long-term temporal dependencies but also makes it difficult for the model to form a consistent understanding of diverse local temporal contexts.
2) In RNN-based networks, U-ConvLSTM performed much better than other backbones. Although these models all utilize recurrent units as their temporal processing module, U-ConvLSTM differs in its architectural design: it is fundamentally a U-Net, with ConvLSTM modules independently inserted into the skip connections between encoder and decoder layers [62]. This strategy reduces the sensitivity to temporal disruptions, thereby enabling the model to perform inference based on spatial information. In contrast, other backbones follow a classical recurrent architecture, where spatiotemporal information is propagated step by step. The missing time steps disrupt the hidden state propagation, making the model update its internal states based on incomplete or fragmented temporal context [65]. As a result, errors accumulate over time, making the spatial information easily influenced by the temporal disturbance, which reduces the reliability of spatial features.
3) In simulated experiments, self-attention-based backbones show superior performance compared to most RNN-based backbones. The time gaps in simulated experiments are more random, leading to a higher frequency of temporal discontinuities. RNN-based architecture relies on sequential information propagation, where historical context is transferred step by step through hidden states. As the frequency of time gaps increases, these propagation pathways are repeatedly disrupted. The gating mechanisms are forced to perform unstable state updates frequently, which makes the network more easily exhibit gradient vanishing and explosion during the backpropagations [66]. Self-attention-based backbones rely on parallel dependency modeling, where temporal relationships are constructed through explicit interactions across all time steps. This allows the model to more easily reconstruct missing information caused by time gaps using strongly correlated observations [67].
4) In contrast, self-attention-based backbones demonstrated inferior performance compared to RNN-based backbones in real-world experiments. As more continuous time gaps appear, the missing regions may lack any observations for interaction, making it extremely difficult to reconstruct information within those time gaps. Due to the strong correlation between adjacent time steps, the mechanism may concentrate attention on the time steps at both ends of the missing segment, which can completely fail the overall temporal modeling. In RNN-based backbones, the propagation of historical information is limited to disrupted, enabling the networks to preserve certain local temporal dependencies. It may support the networks to finish the prediction task.

B. Impact of Each Module on the Proposed Framework

To understand the individual contributions of each module within the proposed method, we conduct an ablation study by systematically removing or replacing specific components.

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TABLE XVI

PERFORMANCE OF DIFFERENT MODULE COMBINATIONS IN SIMULATED EXPERIMENTS ON THE HUNAN SEN DATASET (CL-$F1$: CROPLAND $F1$-SCORE, BG-$F1$: BG $F1$-SCORE, AND M-$F1$: MEAN $F1$-SCORES). MA, KD, FR, AND FS REPRESENT THE MASK, KNOWLEDGE DISTILLATION, FEATURE RECONSTRUCTION, AND FEATURE STANDARDIZATION MODULES, RESPECTIVELY

Module				OA(%)	AA(%)	Kappa	mIoU(%)	M-F1(%)	CL-F1(%)	BG-F1(%)
MA	KD	FR	FS							
×	×	×	×	89.53	87.87	0.71	75.15	85.26	77.34	93.19
✓	×	×	×	89.57	87.88	0.71	75.25	85.34	77.47	93.21
✓	✓	×	×	90.12	88.59	0.72	76.43	86.16	78.76	93.56
✓	✓	✓	×	90.53	89.14	0.74	77.29	86.75	79.67	93.82
✓	✓	✓	✓	90.84	88.12	0.75	78.66	87.72	81.53	93.91

TABLE XVII

PERFORMANCE OF DIFFERENT MODULE COMBINATIONS IN REAL-WORLD EXPERIMENTS ON THE HUNAN SEN DATASET (CL-$F1$: CROPLAND $F1$-SCORE, BG-$F1$: BG $F1$-SCORE, AND M-$F1$: MEAN $F1$-SCORES). MA, KD, FR, AND FS REPRESENT THE MASK, KNOWLEDGE DISTILLATION, FEATURE RECONSTRUCTION, AND FEATURE STANDARDIZATION MODULES, RESPECTIVELY

Module				OA(%)	AA(%)	Kappa	mIoU(%)	M-F1(%)	CL-F1(%)	BG-F1(%)
MA	KD	FR	FS							
×	×	×	×	85.78	83.31	0.41	57.83	69.46	47.14	91.78
✓	×	×	×	88.72	87.38	0.56	66.37	77.80	62.23	93.37
✓	✓	×	×	88.60	84.87	0.58	67.25	78.67	64.11	93.22
✓	✓	✓	×	88.83	88.21	0.56	66.40	77.81	62.16	93.45
✓	✓	✓	✓	89.49	83.98	0.64	71.14	81.99	70.37	93.61

Specifically, the proposed framework mainly consists of four modules as follows.

1) *Mask Module (MA):* Simulates temporal missing environments by producing incomplete SITS through random temporal masks, similar to the strategy used in DA-TD.
2) *Distill Module (KD):* Employs a knowledge distillation framework, where soft supervision signals are provided by a teacher model trained on complete SITS.
3) *FR Module:* Uses the temporal features extracted from complete SITS by the teacher model as supervised signals to guide the student model in learning FR capabilities.
4) *Feature Standardization (FS) Module:* Incorporates standardization steps within the FR Module to mitigate feature distribution shifts caused by time gaps, which prevent the model from overfitting to feature magnitudes. All experiments were conducted in the Hunan SEN dataset, involving both simulated and real-world scenarios on the complete and incomplete subsets. The U-TAE is used as the backbone.

As shown in Table XVI, in the simulated experiments, the model performance exhibits an upward trend as different modules are added. Specifically, the addition of the MA resulted in a slight improvement, with the cropland $F1$ score increasing by 0.17% compared to the baseline without any modules. As discussed in Section IV-B and Appendix A, the benefit of the MA module is partially diminished because simple DA may lead the model to overfit under severe temporal missing conditions. With the addition of the KD module, the performance further improved by 1.67% compared to the setting with only the MA module. This indicates that the soft supervision signal provides information about the relative probabilities of different classes, thereby enhancing the model's generalization capability across varying scenarios. The FR module yielded an improvement of 1.16% in cropland $F1$-score over the MA+KD setting, while the subsequent inclusion of the FS module brought a larger gain of 3.52%. It demonstrates that the standardization helps the model to pay more attention to the global distribution of features rather than local extreme values, thereby improving the model's robustness under varied missing conditions.

As shown in Table XVII, in real-world experiments, the performance trend differs from that observed in the simulated experiments. First, the addition of the MA module yielded a substantial improvement, increasing the cropland $F1$ score by 32.01%. As discussed in Section V-A, this is because the robustness of the self-attention mechanism to temporal missingness becomes ineffective when facing continuous time gaps. In such cases, the benefits introduced by DA far outweigh the potential negative effects caused by overfitting. Second, when only the FR module was added—without the FS module—it even resulted in a performance drop, with the cropland $F1$ score decreasing by 3.04%. This indicates that in real-world scenarios with more continuous time gaps, they may cause substantial distortions in feature magnitudes. Without standardization, forcing the model to mimic the complete temporal feature may lead the model to learn incorrect reconstruction abilities. The addition of the FS module led to a 13.21% improvement in cropland $F1$-score compared to the setting with the other three modules. This further demonstrates the necessity of FS. Additional experiments and analyses under different missing conditions are provided in Appendix B.

C. Analysis of Each Module's Impact via Feature Visualization

To further illustrate the individual contributions of each module in the proposed method, we conduct feature

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Fig. 8. t-SNE visualization of feature maps from different decoder layers in U-TAE. Experiments were performed on the Hunan SEN simulated dataset using nine samples with temporal masking rates of 25%–75%. Baseline (complete) denotes ideal features from the original model under complete SITS, while baseline (incomplete) denotes features from incomplete SITS. MA, KD, FR, and FS represent the mask, knowledge distillation, feature reconstruction, and feature standardization modules, respectively.

visualization by progressively adding different modules. Specifically, we selected spatial–temporal feature maps from different layers of the decoder in the U-TAE for t-SNE visualization. The experiments were conducted on the Hunan SEN simulated dataset, where we selected nine samples with temporal masking rates ranging from 25% to 75%. Baseline (complete) refers to the features extracted by the original model from complete SITS, representing the ideal features that the proposed method aims to reconstruct. Baseline (incomplete) refers to the features extracted by the original model without any modules from incomplete SITS.

As shown in Fig. 8, when only the MA and KD modules are added, the model tends to extract features that focus more on the missing patterns rather than the underlying classes. This effect gradually propagates from deeper to shallower layers, resulting in final feature maps that can distinguish only part of croplands and exhibit substantial intraclass variability. The primary reason is the model's memory limitation: when it pays excessive attention to different masking patterns, the optimization strategy drives it to maximize immediate benefits. Consequently, the model tends to converge to local optima and learn shortcut reasoning, i.e., overfitting to heavily missing conditions, while lacking generalization to other missing conditions and complete SITS.

When the FR module is added, the model's attention to missing patterns is mitigated. However, it still tends to learn values and boundaries of the ideal features in the deeper layers, specifically those located at the edges of clusters. As a result, in the shallower layers, the model still separates the cropland class into two clusters, which has a negligible effect on the model's inference. Once the FS module is added, the model no longer focuses on mimicking the local extreme values of features but instead captures the global feature distribution. This enables the model to extract more discriminative features for class separation, thereby improving its classification performance.

VI. Conclusion

In this article, we propose an FR and prediction joint learning framework for agricultural segmentation tasks using SITS. The framework is designed to enhance the model's robustness under scenarios with missing temporal observations. It can selectively reconstruct key temporal features, thereby enabling the model to obtain reasoning ability on incomplete SITS. Experimental results demonstrate that the proposed framework achieves significant improvements over other state-of-the-art methods across three study areas. More importantly, it exhibits strong generalization capabilities in multiple aspects: 1) generalization across different agricultural SS tasks, including cropland extraction and crop classification; 2) generalization across different sensors, primarily Sentinel-2 and PlanetScope; 3) generalization under different numbers of time gaps, while retaining classification performance on complete SITS; and 4) agnostic across different model architectures, including seven typical backbones. This greatly enhances the applicability of the proposed method, enabling it to be efficiently employed in diverse agricultural monitoring scenarios. In addition, we explored and analyzed the contributions of different modules within the proposed method, which enhanced its interpretability. Although the proposed method is designed for multitemporal agricultural SS tasks, it can also be applied to other "many-to-one" tasks leveraging temporal features. We will further explore its practical utility across different tasks, as well as the model's behavior under various temporal gap patterns and its ability to reconstruct more fine-grained temporal features.

Acknowledgment

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Appendix A
Performance of DA Methods Under Varying Temporal Missing Conditions

The experimental setup is the same as Section IV-C, where simulated experiments are conducted on different datasets under varying temporal missing rates. The missing rate is set from 0% to 25%. As shown in Tables XVIII and XIX, in the cropland extraction task, DA methods exhibit similar or even decreased performance compared to the baseline under complete SITS (0%) and low missing rate (25%) conditions. While DA methods achieve better performance than the baseline under middle (50%) and high (75%) missing rates, they still underperform compared to our method. This demonstrates that the DA methods have overfitted to the middle (50%) and high (75%) missing conditions, and lose their ability to

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TABLE XVIII

RELATIVE INCREASE OR DECREASE IN PERFORMANCE OF OUR AND DA METHODS COMPARED WITH BASELINE METHODS UNDER DIFFERENT TEMPORAL MISSING RATES ON THE HUNAN SEN DATASET (CL-$F1$: CROPLAND $F1$-SCORE, BG-$F1$: BG $F1$-SCORE, AND M-$F1$: MEAN $F1$-SCORES). VALUES IN RED INDICATE A DECREASE COMPARED TO THE BASELINE

Mask ratio	Method	OA(%)	AA(%)	Kappa	mIoU(%)	M-F1(%)	CL-F1(%)	BG-F1(%)
0%	DA-TD	-0.21	0.16	-0.01	-0.78	-0.48	-0.92	-0.10
	DA-WS	-0.60	-0.81	-0.02	-1.45	-0.87	-1.45	-0.37
	Ours	0.50	0.12	0.02	1.51	0.92	1.67	0.28
25%	DA-TD	0.09	0.27	0.00	0.12	0.07	0.08	0.06
	DA-WS	-0.37	-0.88	-0.01	-0.67	-0.40	-0.55	-0.27
	Ours	0.79	0.15	0.03	2.44	1.48	2.74	0.42
50%	DA-TD	0.74	0.81	0.03	1.96	1.21	2.15	0.45
	DA-WS	0.26	-0.60	0.02	1.20	0.77	1.63	0.09
	Ours	1.49	0.51	0.07	4.58	2.82	5.26	0.81
75%	DA-TD	3.16	3.15	0.16	9.23	6.04	11.82	1.77
	DA-WS	2.65	1.09	0.15	8.74	5.80	11.80	1.37
	Ours	4.31	2.50	0.23	13.66	8.78	17.52	2.30

TABLE XIX

RELATIVE INCREASE OR DECREASE IN PERFORMANCE OF OUR AND DA METHODS COMPARED WITH BASELINE METHODS UNDER DIFFERENT TEMPORAL MISSING RATES ON THE HUNAN PLA DATASET (CL-$F1$: CROPLAND $F1$-SCORE, BG-$F1$: BG $F1$-SCORE, AND M-$F1$: MEAN $F1$-SCORES). VALUES IN RED INDICATE A DECREASE COMPARED TO THE BASELINE

Mask ratio	Method	OA(%)	AA(%)	Kappa	mIoU(%)	M-F1(%)	CL-F1(%)	BG-F1(%)
0%	DA-TD	-1.50	-3.10	-0.03	-2.45	-1.40	0.62	-2.02
	DA-WS	-1.29	-2.77	-0.03	-2.09	-1.20	-1.41	-1.00
	Ours	0.22	-0.36	0.01	0.88	0.54	1.09	0.07
25%	DA-TD	-0.86	-2.43	-0.01	-0.98	-0.52	-0.30	-0.73
	DA-WS	-0.87	-2.45	-0.01	-0.98	-0.52	-0.27	-0.74
	Ours	0.60	-0.12	0.03	1.94	1.19	2.21	0.29
50%	DA-TD	1.46	-0.77	0.08	5.19	3.32	6.60	0.64
	DA-WS	1.13	-1.27	0.07	4.50	2.91	5.97	0.42
	Ours	2.60	1.01	0.12	7.69	4.76	8.90	1.40
75%	DA-TD	8.11	3.13	0.74	29.30	22.27	58.81	3.92
	DA-WS	7.42	2.01	0.70	27.45	21.13	56.20	3.51
	Ours	9.32	4.44	0.83	33.10	24.64	64.56	4.59

TABLE XX

RELATIVE INCREASE OR DECREASE IN PERFORMANCE OF OUR AND DA METHODS COMPARED WITH BASELINE METHODS UNDER DIFFERENT TEMPORAL MISSING RATES ON THE FR&CAT S4A DATASET (M-$F1$: MEAN $F1$-SCORES). VALUES IN RED INDICATE A DECREASE COMPARED TO THE BASELINE

Mask ratio	Method	OA(%)	AA(%)	Kappa	mIoU(%)	M-F1(%)
0%	DA-TD	-4.04	1.47	-0.06	-11.20	-7.92
	DA-WS	-8.98	-28.12	-0.15	-31.72	-29.09
	Ours	-0.34	-1.35	0.00	-1.25	-0.86
25%	DA-TD	-3.28	2.83	-0.05	-8.61	-6.20
	DA-WS	-8.40	-27.46	-0.14	-30.66	-28.36
	Ours	0.48	0.30	0.01	2.23	1.39
50%	DA-TD	0.36	11.20	0.01	3.71	1.94
	DA-WS	-5.43	-22.85	-0.10	-24.16	-24.02
	Ours	4.43	7.88	0.08	17.85	11.58
75%	DA-TD	11.39	27.08	0.26	36.88	26.31
	DA-WS	4.28	-11.25	0.10	-2.13	-7.90
	Ours	17.86	23.62	0.39	65.51	45.77

handle complete SITS (0%) and low (25%) missing conditions. In contrast, with the support of reconstructed features, the model achieves better performance in middle (50%) and high (75%) missing conditions compared with DA methods. It also optimized the model's ability to handle complete SITS and keep its robustness under low missing rate (25%) conditions.

As shown in Table XX, in the crop classification task, unlike the DA methods, which suffer a significant performance

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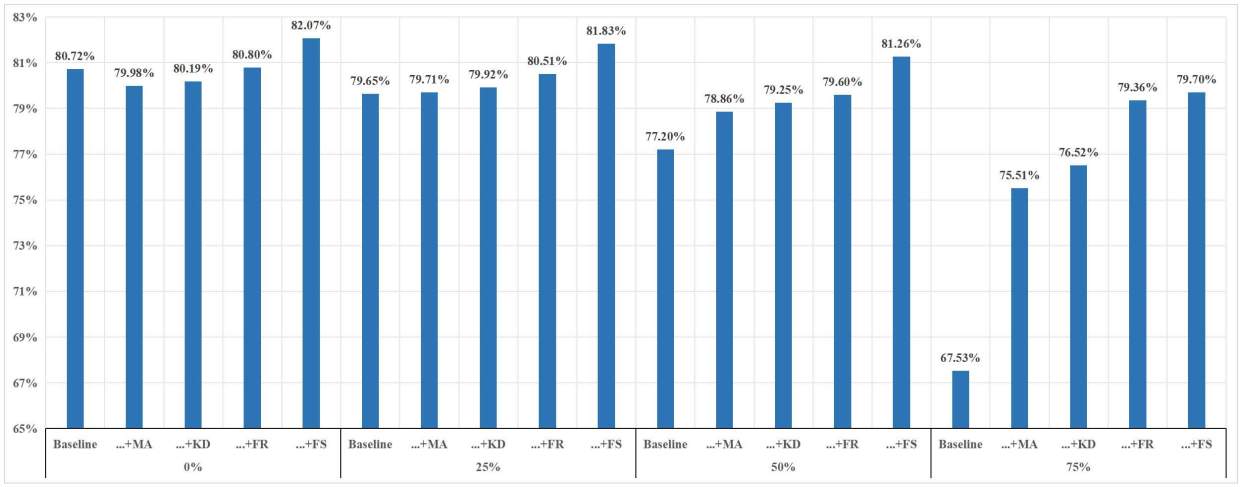

Fig. 9. Performance of different module combinations on cropland $F1$-score under varying temporal missing conditions on Hunan SEN datasets (MA, KD, FR, and FS represent the mask, distill, feature reconstruction, and feature standardization modules, respectively. "..." indicates previously added modules).

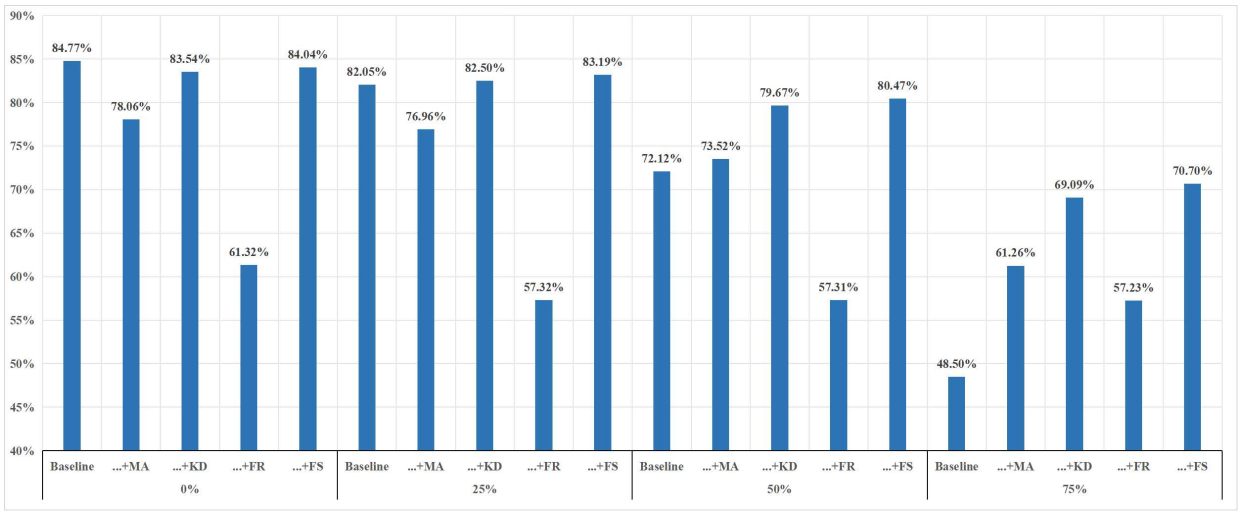

Fig. 10. Performance of different module combinations on mean $F1$-scores under varying temporal missing conditions on Fr&Cat S4A datasets (MA, KD, FR, and FS represent the mask, distill, feature reconstruction, and feature standardization modules, respectively. "..." indicates previously added modules).

decrease on complete SITS, our method shows only a 0.86% drop in mean $F1$-scores. Indicating minimal impact on the model's ability to handle complete temporal inputs. Under the 25% missing rate, our method is the only one that achieves performance improvement over the Baseline, whereas DA-based methods show declines across nearly all metrics. At 50% and 75% missing rates, although the DA-TD method shows improvements in mean $F1$-scores (1.94% and 26.31%, respectively), it is still significantly lower than the improvements achieved by our method under the same conditions (11.58% and 45.77%, respectively).

Appendix B
Effect of Different Module Combinations Under Varying Temporal Missing Conditions

Our exploration of each module remains relatively limited. Therefore, we further investigate the effectiveness of modules under varying levels of temporal missingness. In addition, we conduct similar experiments on the crop classification task. The experiment settings are the same as Section IV-C.

As shown in Fig. 9, with the increase of temporal missingness, the benefits of the MA module also grow. Although it shows a negative effect on complete SITS (0%), it plays a crucial role under more severe missing conditions, especially under high-level (75%) missing conditions. Second, within the 0%–50% temporal missing range, the addition of KD and FR modules yields only limited gains compared to using MA alone. The incorporation of the FS module amplifies the effect of the FR module. However, under the high-level missing condition of 75%, the trend is different. The addition of the FR module exhibits a significant improvement compared to the combination of MA and KD, while the FS module only provides limited gains. This may be because changes in feature magnitude also play an important role under severe temporal missingness, and excessive focus on the feature's distribution might diminish some of the potential benefits. The interaction between FR and FS under different types and levels of temporal missing conditions still requires further investigation.

As shown in Fig. 10, the experimental trend in the crop classification task differs significantly from those observed in

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the cropland classification task. First, without the FS module, the FR module causes severe overfitting in the model, resulting in a loss of classification ability for certain classes. This performance is even worse than using only the MA module under 0%–50% temporal missing conditions. This indicates that, without FS, the reconstructed features may contain significant errors. Second, although the addition of the FR and FS modules improves the model's performance, their gains are smaller compared to those brought by the KD module. This is because the soft supervised signal from the KD module provides the intrinsic relationships among different classes, which significantly reduces the risk of overfitting. We believe that in the crop classification task, alleviating model overfitting may play a more crucial role, while FR also serves as a complementary enhancement.

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